Direction over Size: Sign Dominance in House-Price Expectations*

Elie Chadaida†

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Abstract

How do households form expectations about house prices? This paper uses two household surveys to show that households form their short-term expectations of house prices mainly in response to the sign of perceived past price growth rather than its size. When the perceived direction of prices turns, expectations jump, because households switch from expecting falls or constant prices to expecting rises. It then studies the consequences of such expectations in a partial-equilibrium life-cycle model with a rent-or-own decision, in which households hold the beliefs measured in the surveys. Because they wait for prices to turn before expecting them to rise, households delay their purchases, and purchases bunch when prices turn.

Keywords: house-price expectations, perceptions, coarse beliefs, housing tenure.
JEL codes: D84, E71, G51, R21, R31.

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†Centre for Macroeconomics, London School of Economics.

1 Introduction

Housing represents a significant share of household wealth. In the 2022 Survey of Consumer Finances, two-thirds of families owned their primary residence, and primary residences were the largest category of household assets (Aladangady et al., 2023). Decisions about whether to buy or keep renting, when to move and how much to borrow therefore depend on households' housing needs, incomes and access to credit, but also on what they expect to happen to house prices: a household that buys a home also takes a large, typically mortgage-financed position in a single asset.

A large body of survey and experimental evidence studies these expectations. It shows that households extrapolate their expectations from past price changes (Case et al., 2012; Kuchler and Zafar, 2019; Kuchler et al., 2023). It also shows that these expectations move behavior. In the housing market, households expecting higher price growth are more likely to buy (Bailey et al., 2018), invest more in a housing-linked asset (Armona et al., 2019), and delay selling their homes (Bottan and Perez-Truglia, 2020), while beliefs about future prices also shape mortgage leverage (Bailey et al., 2019). Beyond housing, these expectations affect consumption and saving: renters with higher house-price beliefs cut their spending, consistent with saving for a costlier future purchase, while homeowners' spending barely responds (Chopra et al., 2025). This contrasts with realized price changes, which raise homeowners' spending through a housing-wealth effect approximately equal to the marginal propensity to consume times the change in housing wealth (Berger et al., 2018).

Expectations played a key role in the housing boom and bust of the 2000s. Shifts in beliefs about housing demand and prices are the main driver of house-price movements in Kaplan et al. (2020), although the relative role of credit conditions remains debated (Landvoigt et al., 2015). Burnside et al. (2016) show that a model in which households hold different views about long-run fundamentals, which spread through social interactions, can produce housing cycles of different shapes. Chodorow-Reich et al. (2024) show that beliefs and over-optimism can amplify booms. The data also display features such as short-run momentum in house prices (Glaeser et al., 2014), which models with extrapolative beliefs help deliver (Glaeser and Nathanson, 2017). Expectations also shaped the decisions that led to the crisis: many choices by borrowers, lenders and investors that look bad ex post were reasonable ex ante given the optimistic price expectations widely shared at the time (Foote et al., 2012), which is why the crisis has been described as a crisis of beliefs (Gennaioli and Shleifer, 2018). Understanding how house-price expectations are formed is therefore important, both for understanding households' financial decisions and for understanding housing cycles.

Existing empirical studies typically summarize extrapolation with a linear specification (see Kuchler et al., 2023, for a review). If expectations instead respond to the *direction* of past changes, beyond their size, this approach would miss part of the picture: a linear specification would attribute a jump at zero to a slope. In this paper, I ask which feature of past house-price growth forecasts build on. I show that house-price expectations respond to the direction households perceive in past local prices, beyond the size of the change. I call this pattern *sign dominance*. I then build a partial-equilibrium life-cycle model of the rent-or-own decision whose beliefs are taken directly from the

survey, and use it to study what sign dominance implies for households. My contributions are threefold.

First, I show that households' short-term house-price expectations jump at the sign of past price growth as they perceive it, and respond much less to its size.[1] The main evidence comes from the New York Fed's SCE Housing Survey, which asks households for the value of a typical home in their ZIP code a year ago, today and a year from now. In this survey, two respondents interviewed in the same state and month, one of whom perceives that local prices fell slightly and the other that they rose slightly, differ in their expectations for the coming year by 2.8 percentage points. By contrast, two respondents whose perceived rises differ by ten points differ in their expectations by only 1.9 points. The perceived local direction also moves respondents' expectations for prices in the whole country, again through its sign far more than its size, in line with evidence that households extrapolate local experience into their expectations about the nation (Kuchler and Zafar, 2019). The step is robust to standard demographic controls, to local market conditions, to macroeconomic conditions, which the comparison within a survey year holds fixed, to respondents' other expectations, and to a set of checks on how respondents answer the survey.[2]

Second, I explain how the step works. I decompose the jump into an *extensive margin* contribution — due to households switching from expecting a decline or no change to expecting an increase — and an *intensive margin* contribution — from households revising their expectations within the same direction — and show that the extensive margin accounts for about 90 percent of the observed jump in the SCE.[3] In the spirit of Carlson and Parkin (1975), I build a model of the answers in which a latent belief that lies inside a band is reported as "no change", and show that this step is in the belief, not only in how it is reported. The Michigan Surveys of Consumers, which add answer categories (they ask whether households expect prices to go up, stay the same or go down, and then by how much), show the same pattern over the national cycle. This survey also adds a panel of households re-interviewed six months later. Within a household, answers switch category gradually after the national sign turns, but households re-interviewed in the same answer category barely revise their expectation. However, the Michigan national series has an important limitation. It identifies the step from the national cycle, with one bust and few changes of sign, and inflation controls reduce the national step by a third to three-fifths; it is no longer statistically significant once inflation excluding shelter is controlled for. I report this limitation in full and base the main result on the SCE. The SCE also records housing tenure: renters' step is about twice owners', although the difference is imprecisely estimated.

Third, I study the implications of sign dominance in expectations, using a simple two-period model that I then extend into a life-cycle model. In the two-period model, a homeowner who

[1] The pattern concerns the direction households *perceive*. In 2014–24 few states saw actual price declines, so the step at the sign of actual local growth cannot be estimated precisely in the SCE; Section 2.4 discusses how perceived and actual price changes differ.

[2] The checks on answers are the following. The step survives dropping answers that are mechanically consistent across the three dollar values; it is larger in the years in which the survey asks for the past change directly; it survives excluding round numbers; and it cannot come from an explicit "no change" option, which the SCE does not offer. Placed at other thresholds, the step is also largest at zero. Section 3.5 presents all checks and Appendix C reports them in full.

[3] The decomposition rebuilds the mean from separate fits of the shares and the amounts, so the jump it decomposes differs slightly from the step estimated directly on the mean; Section 3.7 explains.

expects prices to grow faster (slower) than they actually do consumes more (less) at time 0, because her expected lifetime budget is higher (lower). A renter, on the other hand, buys when she expects owning for a year to cost less than renting, i.e. when the interest forgone on the price, net of expected appreciation, falls below the rent. This is the user-cost condition of Poterba (1984) and Himmelberg et al. (2005). Since expectations step at zero, the renters near their buying threshold buy when prices turn, increasing the share of homeowners. Since I do not derive an analytical solution for the density of renters near the threshold, I extend this mechanism into a life-cycle model with income risk, a down payment and a costly rent-or-own choice, in the tradition of Berger et al. (2018) and Kaplan et al. (2020). A distinguishing feature of the model is that its belief rule is not chosen: households expect 0.4 percent nominal growth after a fall in prices and 3.3 percent after a rise, the values SCE respondents expect. Along the actual 1998–2025 national price path, a household with such expectations buys most in 2012, the year its belief turns, while a household with full information buys a year earlier, at the trough. The two-period formula holds closely: 17.8 percent of renters have buying thresholds between the two beliefs, and purchases at the turn increase by 15.8 percentage points. The wave needs a step of the size measured, however: with the step at the low end of its confidence interval, it largely disappears. Households with sign-dominant expectations who have enough liquid wealth to buy a house prefer to rent for one extra year, compared with a similar household with full information.

These results speak to several broader questions. For the measurement of expectations, they show that the published Michigan median expectation appears anchored at zero when national prices fall, because most households then expect prices to "stay about the same". For models of housing cycles, they suggest that switches in beliefs occur when perceived price growth changes sign. On the policy side, more transparent information on local prices could lead to more accurate perceptions of past price growth, and thus to better-informed expectations.

Related Literature

House-price expectations. This paper relates to the literature on house-price expectations, reviewed by Kuchler et al. (2023). This literature shows that households extrapolate past price changes, whether actual, locally experienced or perceived, and that local experience also shapes their expectations about the nation (Case et al., 2012; Kuchler and Zafar, 2019; De Stefani, 2021; Kiesl-Reiter et al., 2025); that households revise their expectations when informed about past prices (Armona et al., 2019); and that expectations under-react to past growth in the short run, over-extrapolate it in the long run and underestimate the reversion of prices to fundamentals (Case et al., 2012; Armona et al., 2019; De Stefani, 2021; Ma, 2020). These studies measure how strongly expectations respond to past price growth. This paper contributes to this literature by disentangling which features of past price growth expectations respond to: I separate a step at the perceived sign of past growth from the response to its size, and decompose it into an extensive margin and an intensive margin in the aggregate expectation.

Measuring expectations. I turn qualitative answer categories into quantitative expectations by adapting the band framework of Carlson and Parkin (1975), in which a belief is reported as "no change" when it lies close to zero. My interpretation of the expectations measured, and of their impact on my results, draws on studies showing that round and zero answers can reflect

uncertainty (Binder, 2017; Manski and Molinari, 2010). Andre et al. (2026) also warn that a significant share of respondents have not formed a house-price expectation before being asked in a survey (only 41–45 percent have), which could undermine the reliability of survey measures of house-price expectations. This work guides my robustness checks: I show that the step also appears in a survey that elicits dollar values and offers no "no change" option, and that it survives excluding round answers or reading them as intervals.

Beliefs in housing models. Quantitative housing models differ in how beliefs are formed and evolve: beliefs extrapolate past prices smoothly, switch between regimes for exogenous reasons, or spread through social interactions (Glaeser and Nathanson, 2017; Kaplan et al., 2020; Burnside et al., 2016), while recent models derive them from survey data (Ludwig et al., 2026; Adam et al., 2025). In a model estimated on housing portfolio choices, Landvoigt (2017) finds that expected appreciation mainly drives the rent-or-own margin. I study, in a partial-equilibrium life-cycle model in the tradition of Berger et al. (2018) and Kaplan et al. (2020), the implications of a belief rule measured in the SCE whose shape is a step rather than a slope, a shape that, to my knowledge, has not been studied in such models.

Coarse beliefs and expectation formation. Finally, this paper also speaks to the literature on expectations departing from full information. Models of categorical thinking (Mullainathan, 2002), sparse attention (Gabaix, 2014) and information-constrained choice (Sims, 2003; Matějka and McKay, 2015; Jung et al., 2019) predict that agents respond to coarse features of their environment, while sticky information (Mankiw and Reis, 2002; Carroll, 2003) and costly updating (Reis, 2006; Alvarez et al., 2011) predict gradual adjustment. The expectations I measure broadly fit this view: they respond to perceived rather than actual past prices, mainly to one dimension of these perceptions, their sign rather than their size, and they adjust gradually after a turn.

2 Data and Measurement

This section describes the two surveys, the house-price data and the three measures of past growth used throughout the paper. Because the main evidence concerns perceived rather than actual prices, I end this section by documenting how households' perceptions relate to actual price changes.

2.1 The SCE Housing Survey

The New York Fed's Survey of Consumer Expectations (SCE) includes a housing module each February (Armantier et al., 2017). In 2014–20, owners and renters report the value of a typical home in their ZIP code today, a year ago and a year from now; the perceived past change and the expected change are computed from these values. From 2022 onwards, respondents report the past change directly. The 2021 module does not ask for these values, so the SCE sample covers ten years: 2014–20 and 2022–24. The module never asks whether prices will go up, stay the same or go down, so it has no explicit "no change" option. The public housing module carries no geography. I link each respondent to the core survey through the respondent identifier, which provides the state of residence, demographics and the respondent's other expectations, including

the expected change in national home prices. The main sample has 8,028 respondent-years, of whom 740 perceive a local decline; 17.4 percent expect exactly zero change.

Two features of the SCE are important for what follows. First, no respondent appears in two February modules, so the survey has no within-person dimension. Second, all respondents in a given year are interviewed in the same month. The identifying variation for the step is therefore *across respondents surveyed in the same year*: inflation, interest rates and every other national condition are common to all of them.

2.2 The Michigan Surveys of Consumers

The Michigan survey has asked homeowners about future house prices since 2007 (Survey Research Center, University of Michigan, 2025). The question has two steps. Respondents first say whether prices of "homes like yours in your community" will go up, stay about the same, or go down over the next twelve months; those who answer up or down then give a percentage. Answers of "stay about the same" are coded as zero. The estimation sample has 89,676 respondent-months from 2007 to 2024 (it ends in 2024 because it also uses realized growth over the following year), and a subset of 32,298 pairs of respondents is re-interviewed six months later. The public data identify only the respondent's Census region. The identifying variation in Michigan is therefore the *national cycle over time*: one bust, 2007–12, and 4 changes in the sign of national growth. What Michigan adds relative to the SCE is the qualitative answer and the same households observed twice.

2.3 House Prices and Three Measures of Past Growth

Past price growth s_t is the latest twelve-month growth that was known at the interview date, allowing for publication lags. The national series is the S&P CoreLogic Case-Shiller index. Regional growth is computed from Zillow metro indices: twelve-month growth within each metro area observed at both dates, averaged across metros in each Census region.[4] State growth is computed in the same way from county indices. Realized future growth g'_t is growth over the twelve months after the interview, on the same geography as s_t.

Table 1 defines the three measures of past growth used throughout. All results come from a single run that applies the same sample rules and the same inference to the three measures, and I report every result for the SCE and Michigan side by side. Table 2 reports summary statistics.

Table 1: Three measures of past growth

Label	Past growth s	realized growth g'	Variation used
SCE, perceived local change	the respondent's perceived past-year change in her ZIP code	next-year growth in her state	across respondents within a year
Michigan, national growth	lagged twelve-month Case-Shiller, national	next twelve months, national	over time (one bust)
Michigan, regional growth	lagged twelve-month Zillow, Census region	next twelve months, same region	over time and across four regions

[4] Averaging price *levels* across the metros available in each month would mix price changes with changes in coverage, which grew sharply in 2008–12. Appendix A gives the details.

Table 2: Summary statistics

	N	Mean	SD	P10	Median	P90
Panel A: Michigan — respondent-months with a house-price expectation, 2007–2025						
Expected house-price change, 1y (%)	90,430	1.44	6.35	-3.00	0.00	8.00
National growth published at the interview, Case-Shiller (%)	90,430	3.48	7.03	-5.40	4.50	10.72
Lagged regional growth, per-MSA series (%)	90,430	3.32	5.80	-3.94	3.61	9.97
Expected income change, 1y (%)	86,199	1.49	11.36	-7.00	1.00	10.00
Own experienced house-price change, ann. (%)	28,567	2.31	30.03	-33.33	0.00	40.00
Revision across the re-interview (%)	32,298	-0.07	7.07	-6.00	0.00	6.00
Age (years)	89,301	55.63	15.85	33.00	57.00	75.00
Panel B: Michigan — monthly series, months with a published expectation, 2008–2025						
Case-Shiller growth, 12-month (%)	216	3.64	6.99	-5.79	4.48	10.71
Median expected house-price change (%)	216	0.53	0.78	0.00	0.30	1.55
Mean expected house-price change (%)	216	1.40	1.41	-0.60	1.80	2.90
Share expecting an increase (%)	216	38.45	13.73	18.00	42.00	54.00
Share expecting a decrease (%)	216	13.04	6.19	7.00	11.00	23.00
Forecast error, median − realised (%)	204	-3.82	6.18	-10.58	-4.10	3.70
Panel C: SCE Housing Survey — respondent-years in the main sample, 2014–2024						
Expected local house-price change, 1y (%)	8,028	4.31	8.11	0.00	3.23	12.50
Perceived past-year local change (%)	8,028	6.15	9.85	0.00	4.17	18.18
Expected own-home change, 1y (%)	5,800	2.93	5.62	0.00	2.26	8.33
Actual state growth, past year (%)	8,028	5.31	4.23	1.52	4.52	9.32
Actual state growth, next year (%)	8,028	4.96	3.24	1.33	4.65	8.81
Owner (share)	8,028	0.74	0.44	0.00	1.00	1.00

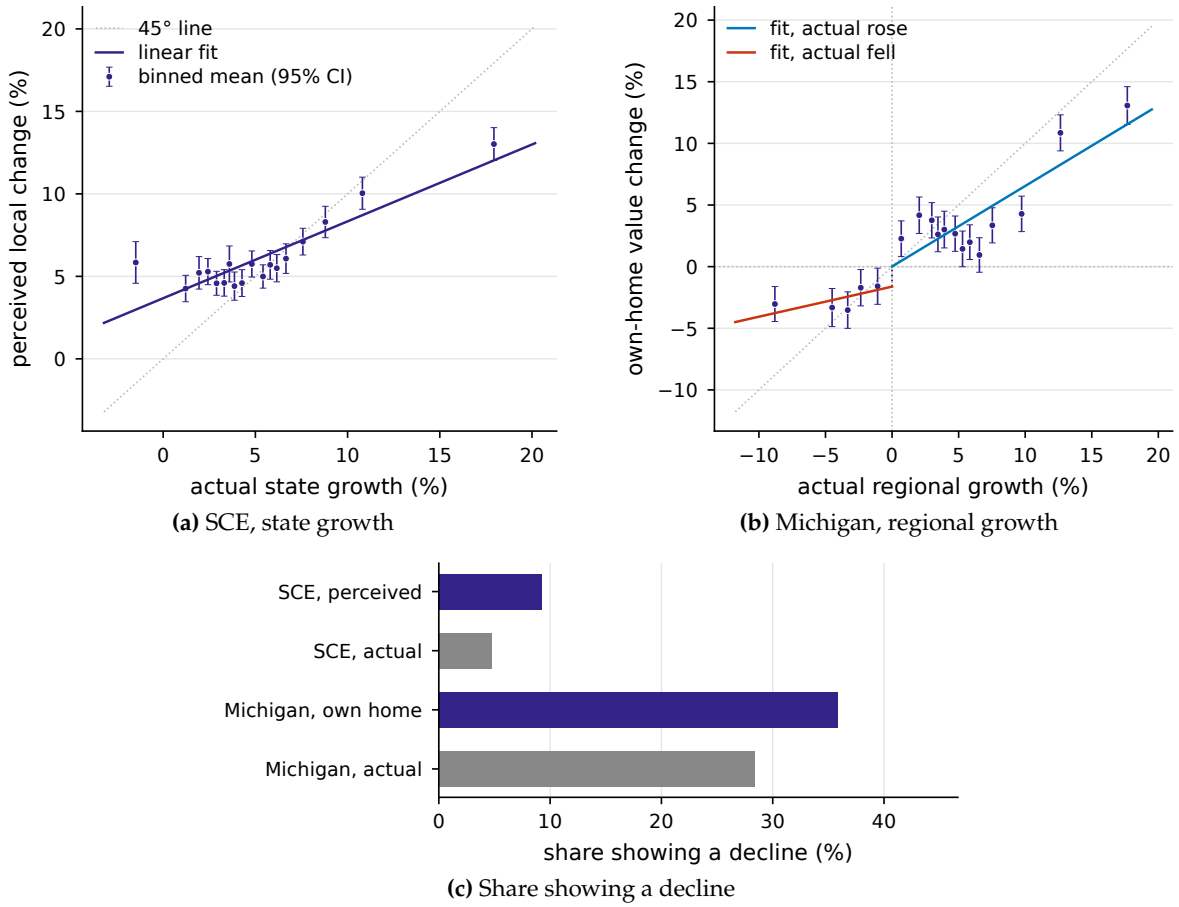
Notes: Panel A: Michigan Surveys of Consumers, respondent-months with a one-year house-price expectation (asked of homeowners only). Lagged growth is the latest 12-month growth known at the interview: national Case-Shiller, and regional growth computed per metro area and averaged within Census region. Own experienced change: annualised change in the respondent's reported home value across the re-interview, $|x| < 100$. Revision: second minus first expectation, re-interview pairs. Panel B: monthly published Michigan series in months with a median expectation and lagged Case-Shiller growth. Panel C: SCE Housing Survey respondents with an expectation, a perceived past change, a state, a survey year and next-year state growth (the main SCE sample, Table 3); state growth from Zillow.

2.4 What Households Think Happened

Figure 1 plots perceived against actual past price changes. Three features stand out. First, perceptions are *compressed*: regressing the perceived past change on the actual change gives a slope of 0.40 in the SCE with year effects, and 0.63 in Michigan, where the perceived change is the change in a respondent's reported home value across the re-interview. A slope of one, as under full information, is rejected under every standard error (Appendix Table B.1). Second, perceptions are *dispersed*: respondents in the same state and year perceive different directions, which is precisely the variation the SCE uses. Third, perceptions are often *wrong in sign*. In the SCE sample, 740 respondents perceived a local decline, while only 385 lived in a state where prices actually fell. The third feature shapes the interpretation of what follows. The step can only be measured at the perceived sign, and the paper is about how households react to the direction they *believe* prices took.[5] Large errors in perceived past local price growth are also documented by Armona et al.

[5]Many perceived declines are therefore errors. Classical noise in a running variable does not create a discontinuity; it smooths a true one (Pei and Shen, 2017; Davezies and Le Barbanchon, 2017). Noise correlated with expectations, such

(2019) and Kiesl-Reiter et al. (2025).



(a) SCE, state growth

(b) Michigan, regional growth

(c) Share showing a decline

Figure 1: Perceived against actual past price changes. Binned means of the perceived past-year change against the actual change in the respondent's state (SCE) or region (Michigan), with the fitted line and the 45-degree line implied by full information.

3 Sign Dominance in House-Price Expectations

This section presents the empirical evidence. I start from a puzzle in the published Michigan median series, which motivates looking at the full distribution of answers. I then set out the specification, document the step, and examine whether it is really about direction. Finally, I show how the step works.

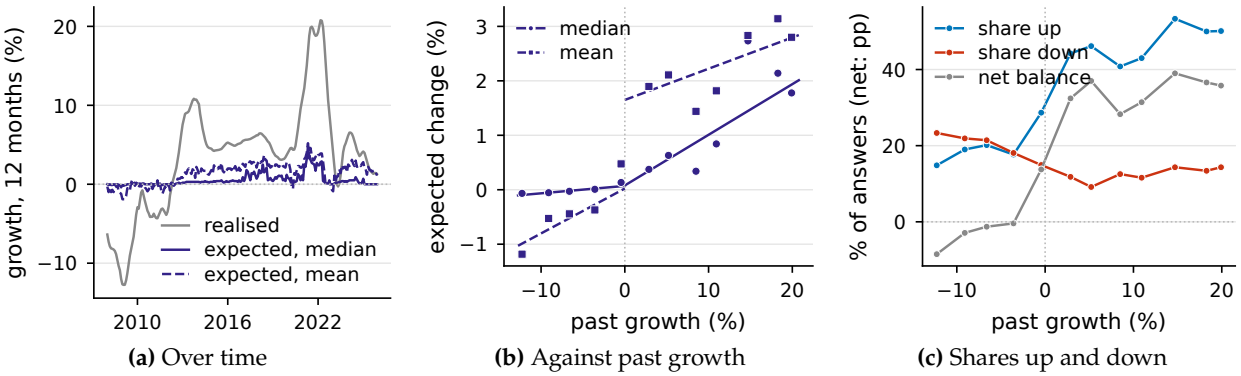
3.1 The Floor in the Median

At first glance, the published Michigan median expectation rises with past gains but is flat after declines (Figure 2): with fits that allow a step at zero, it rises by 0.093 per point of past gains and by 0.014 per point of past declines. Read literally, this says that households extrapolate gains but never expect prices to fall, a floor like the one documented for inflation expectations by Gorodnichenko and Sergeyev (2021). The mean does not show this pattern: it jumps by 1.63 points at zero, and it responds about as much to past declines as to past gains (0.082 and 0.057 per point; $p = 0.623$ for equal slopes).

as general pessimism, is addressed by the mood placebo in Section 3.5.

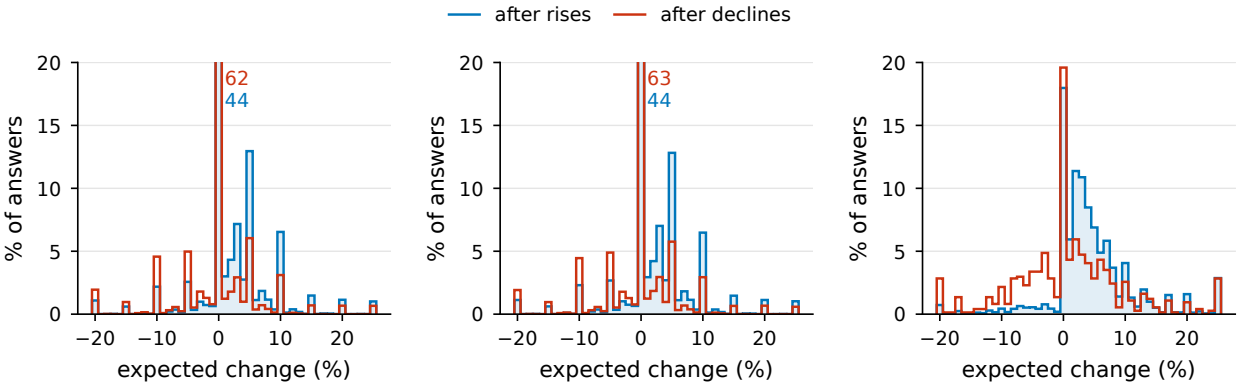
This discrepancy between the mean and the median calls for a look at the distribution of answers (Figure 3). In Michigan, 49.0 percent of answers are exactly zero, that is, "stay about the same": 61.7 percent after national declines and 44.1 percent after rises. The median is zero whenever fewer than half of respondents answer "up" and fewer than half answer "down". After declines, most respondents answer "same", so the median is zero regardless of how the "down" share moves. The floor is therefore a property of the median, not of individual expectations, and it does not require many "no change" answers. In the SCE, where expectations are computed from dollar values, only 17.4 percent of answers are exactly zero, yet the median is also zero after a perceived decline and does not move with its size. A perceived decline moves the lower half of the distribution below zero: 34.9 percent of respondents expect a fall, a quarter expect a fall of at least 3.6 percent, and the mean falls to 0.89 percent. But 46.1 percent still expect a rise, so neither direction reaches half, and the median lands on the 19.1 percent who expect no change.

The puzzle points to the object of interest. What moves when past growth changes sign is not so much the size of the expected change as its direction: which answer category households choose. The rest of this section makes this precise.



(a) Over time **(b)** Against past growth **(c)** Shares up and down

Figure 2: The median and the mean. (a) Published Michigan median and mean expected twelve-month house-price change, and realized Case-Shiller growth. (b) Binned means against lagged national growth, with fits that allow a step at zero. (c) Shares expecting an increase and a decrease.



(a) Michigan, national growth **(b)** Michigan, regional growth **(c)** SCE, perceived local change

Figure 3: The distribution of answers, in one-point bins, by the sign of past growth: Michigan, by the sign of lagged national growth; Michigan, by the sign of lagged regional growth; SCE, by the sign of the perceived past-year change. Tails pooled at −20 and +25.

3.2 Specification and Inference

To separate the response to the direction of past price changes from the response to their size, I estimate

$$y_{it} = a + J\,\mathbf{1}[s_t \geq 0] + b^- s_t^- + b^+ s_t^+ + \varepsilon_{it}, \qquad s^- \equiv \min(s, 0), \quad s^+ \equiv \max(s, 0), \tag{1}$$

where y_{it} is respondent i's expected twelve-month price change. The step J measures how much expectations change when past growth crosses zero, and the slopes b^- and b^+ measure the response to the size of past declines and past gains.

A linear regression imposes $J = 0$ and $b^- = b^+$, and would attribute a jump at zero to a slope. To see this, suppose households expect zero growth after any decline and two percent after any rise; a linear regression on s_t finds a positive slope, whereas (1) finds $J = 2$ and no slope at all.

In the SCE, the regression includes survey-year effects, which absorb everything common to a February survey, and a separate step J_0 for perceived changes of exactly zero; both J and J_0 are measured relative to a perceived decline. In Michigan, national growth is common to all respondents in a month, so the baseline specification has no controls, and controls enter as robustness checks. Standard errors are clustered by state in the SCE and by calendar year in Michigan, and p-values come from a wild cluster bootstrap by year.[6]

3.3 The Step

Table 3 and Figure 4 report the estimates. In the SCE, respondents who perceive a slight local decline expect 2.815 points less growth than respondents in the same year who perceive a slight rise ($p = 0.005$).[7] The slopes, on the other hand, are modest. A perceived rise ten points larger raises expected growth by only 1.86 points, less than the step; the downside slope is -0.102 and insignificant: below zero, the size of the perceived fall hardly matters.

Michigan shows the same shape over the national cycle, smaller in size: expectations step by 1.284 at zero on national growth ($p = 0.006$) and by 1.766 on regional growth, and the slopes are small. Because Michigan identifies these estimates from the national cycle, they are more exposed to national conditions than the SCE; Section 3.5 shows which of them survive controls for these conditions.

To be precise, the claim is not that the size of past changes is irrelevant: for large moves, the slopes add up. The claim is that expectations jump at zero, which a linear specification cannot capture, on top of a modest response to size, and that the jump is large relative to the response to the moves households typically perceive.

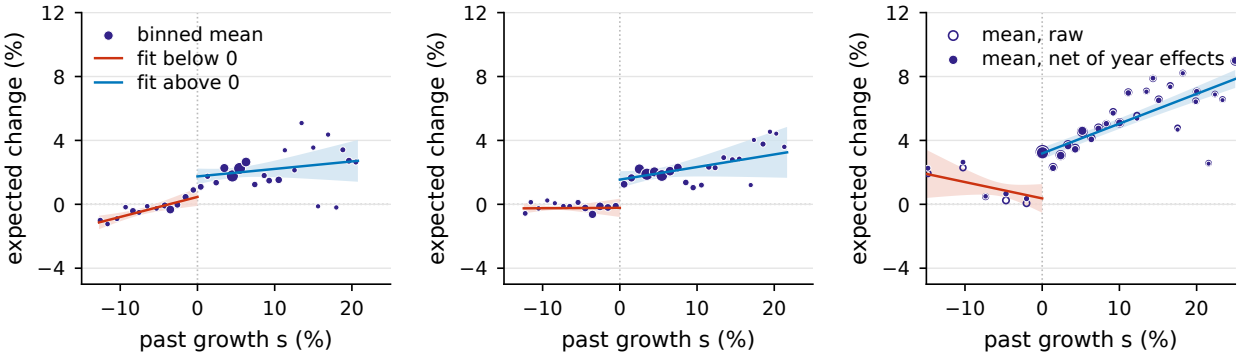
[6] With only 10 to 18 years, the usual cluster-robust test over-rejects in simulations calibrated to these data, while the wild cluster bootstrap over-rejects much less (Cameron et al., 2008; MacKinnon and Webb, 2017; Webb, 2023). Appendix B reports the simulations and the estimates under alternative standard errors.

[7] Respondents who perceive exactly no change have their own intercept in (1). They expect about as much growth as those who perceive a rise ($J_0 = 2.945$), so the step separates "prices fell" from "prices did not fall".

Table 3: The step at the sign of past growth: regression (1)

	SCE	Michigan	
	Perceived local change	National growth	Regional growth
Step J	2.815*** (0.506)	1.284*** (0.373)	1.766** (0.413)
Step at exactly zero, J_0	2.945*** (0.547)	–	–
Slope b^-	−0.102 (0.076)	0.125* (0.036)	0.002 (0.035)
Slope b^+	0.186*** (0.013)	0.047 (0.041)	0.079 (0.047)
Year effects	yes	no	no
N	8,028	89,676	89,676

Notes: SCE: expectation on the perceived past-year local change, 2014–2024; J (perceived rise) and J_0 (no change) are relative to a perceived decline. Michigan: expectation on national Case-Shiller growth published at the interview, or on Census-region growth, 2007–2024. Standard errors clustered by state (SCE) or year (Michigan). Stars: wild cluster bootstrap p-values by year (restricted, Webb weights; SCE: all 6^{10} Webb sign draws enumerated; Michigan: 1,999 draws): $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



(a) Michigan, national growth **(b)** Michigan, regional growth **(c)** SCE, perceived local change

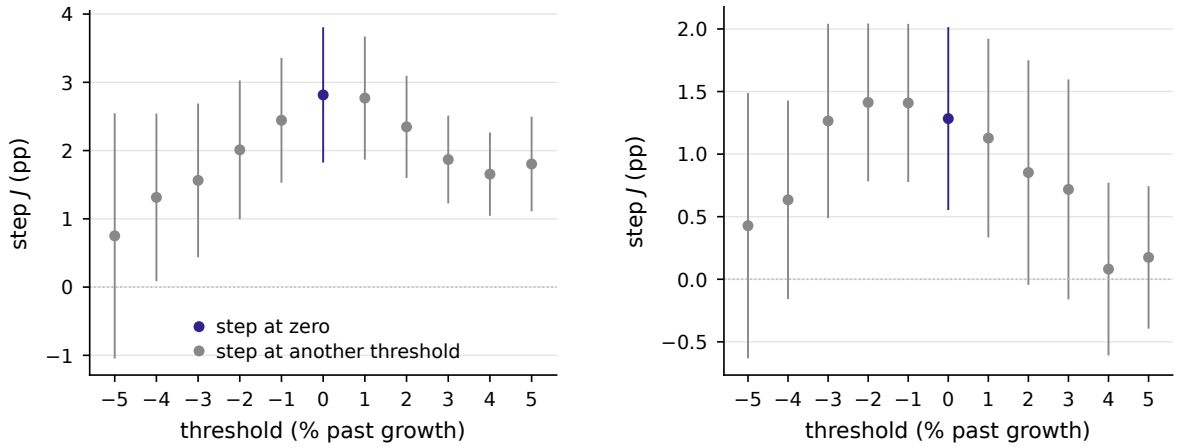
Figure 4: The step in the mean. Binned individual expectations against past growth, with the fit of (1) and 95% bands, on the same axes in every panel: Michigan, lagged national growth; Michigan, lagged regional growth; SCE, perceived past-year local change, net of year effects.

3.4 Is the Step at Zero?

If expectations respond to the direction of past changes, zero is special: it is the only point at which the direction changes. A step of similar size at, say, +3 or −2 percent would instead suggest that households respond to large against small changes, or that (1) picks up some other nonlinearity. I therefore place the step at thresholds from −5 to +5 percent growth, one at a time, and then let the data choose its location with a threshold regression (Hansen, 2000), with a bootstrap interval for the location (Figure 5).

In the SCE, the step is largest at zero and fades away from it, more slowly above zero than below, and the estimated location lies a little above zero, with an interval that contains zero. The Michigan national series cannot locate the step: steps of similar size appear anywhere from −3 to +1.[8] Since the SCE, which carries the main results, locates the step at zero and the national series cannot locate it, I keep the threshold at zero in both surveys.

[8]This is a limit of the national data rather than evidence against direction: only 22 of 216 months lie within two points of zero, so thresholds near zero split the months almost identically.



(a) SCE, perceived local change **(b)** Michigan, national growth

Figure 5: Placebo thresholds. The step in (1) placed at thresholds from −5 to +5 percent growth, one at a time, with 95% intervals: SCE, perceived local change, and Michigan, national growth.

3.5 Is the Step Really About Direction?

A step at the perceived sign could arise for reasons other than a response to direction. I consider six alternatives in turn; Appendix C reports every check in full.

Inflation and the macro cycle. While the year effects in the SCE absorb inflation, mortgage rates and every other national condition, the Michigan step is identified from the national cycle and is sensitive to inflation: inflation controls reduce it by a third to three-fifths, and it is no longer significant once inflation excluding shelter is controlled for.[9] With a single bust, the national series cannot separate the step from inflation and general sentiment, which is why the main results rest on the SCE.

Different households, or different local conditions. Respondents who perceive a decline may differ systematically from those who perceive a rise, or may live in different markets. With demographic controls in both surveys (age and its square, sex, education, and region or state), the step remains. In the SCE, it also survives state and state-by-year fixed effects, which compare respondents in the same local market and the same year who perceive different directions.[10]

General optimism. A pessimistic respondent may report both that prices fell and that they will fall. If general optimism drove the step, her other expectations would step too. They step much less: expectations for gold and gas prices do not step, stock-price expectations step by 0.140 standard deviations, against 0.413 for house prices, and expectations of inflation and unemployment step by about half as much as house-price expectations. Controlling for expectations of stock prices, gold, gas, inflation and unemployment leaves 93 percent of the house-price step (Table C.1).

[9]The step remains significant with the median inflation expectation and, at the 10 percent level, with headline inflation. Excluding shelter, whose cost follows house prices, weakens the step rather than strengthening it, so the drop is not a bad-control effect through rents. Netting each respondent's own inflation expectation out of her house-price expectation gives a larger step, located at nominal rather than real zero, but about half of it comes from inflation expectations falling when house prices turn up, and it is not significant with inflation excluding shelter as a control either. Appendix D reports every specification.

[10]The controls enter the regression directly, which is equivalent to residualizing both the outcome and the regressors on them, as in Wang (2023). I do not residualize the perceived change itself, because its zero defines the direction.

Consistency between answers. In 2014–20, SCE respondents report three dollar values: the price of a home one year ago, its current price, and its expected price one year ahead. Some respondents may anchor their answers on today's price, reporting past and future prices at equal distances from it. This mechanically produces symmetric perceived and expected price changes, without necessarily reflecting expectations responding to the perceived direction of prices. Excluding answers in which today's price is exactly midway between the past and future values, the step remains 2.626 ($p = 0.007$). In 2022–24, when the past price change is asked directly and this form of anchoring cannot generate symmetric answers, the step is 3.331, larger but imprecise with only three years.

The question format and round numbers. A step could arise from Michigan's up, same, down question alone. The SCE has no such question, and the step appears in the sign of the implied change. Many answers are round numbers, which could mismeasure the amounts. The step survives excluding round answers, smaller but significant in every measure, and in the band model of Section 3.7, reading round answers as intervals leaves the latent step unchanged.

The actual direction. One would like to know whether expectations also step at the sign of *actual* local growth. In the SCE this cannot be tested: only 385 respondents faced an actual decline in 2014–24, and the first stage of an instrumental-variable version is weak (Appendix Table C.1). In Michigan, comparing regions within the same survey month, the regional step is -0.221 ($p = 0.087$): households track the national sign rather than their region's. Census regions are coarse, however: with metropolitan identifiers, De Stefani (2021) finds that expectations vary with local price growth, and obtaining such identifiers is a natural next step for this research.

3.6 National Expectations

The core SCE also asks each month for the expected change in national home prices over the next year. Therefore, for housing-module respondents I observe the perceived local change and the national expectation in the same February. National expectations show the same step as local ones (Appendix Table C.1): at the perceived local sign, they step by 2.85 ($p = 0.003$), against 2.77[11] for the local expectation on the same respondents, and they respond even less to the size of the perceived change. Households thus carry the direction of their local experience over to their expectations for the whole country, in line with Kuchler and Zafar (2019).

Two readings are consistent with this result: the perceived direction works as a general signal about the housing market, or it reflects housing-specific pessimism. General mood is ruled out by the placebo above; housing-specific pessimism is not, and I do not attempt to distinguish the two.

3.7 How the Step Works

Having documented the step and its properties, I now turn to how it arises. When households perceive a rise in prices, do they expect larger increases, or do they switch from expecting falls to expecting increases? And is the step in their beliefs themselves, or only in how they report them?

[11] This is the step of Section 3.3, re-estimated on the 7,170 respondents who also answer the national question.

3.7.1 Switching Direction, Not Amounts

In the SCE, a respondent's expected change is a rise, a fall, or exactly zero; in Michigan, respondents answer up, down or same, and give the amount in a subsequent question. Answers of no change are zero, so the mean expectation is the share of respondents expecting each direction times the average change they expect,

$$\mathbb{E}[y \mid s] = \sum_{k \in \{u,\, d\}} P_k(s)\, m_k(s),$$

where u denotes a rise and d a fall. Let $P_k^{\pm}$ and $m_k^{\pm}$ be the fitted values just above and just below zero. Adding and subtracting $P_k^{+} m_k^{-}$ splits the jump in the mean exactly into two parts:

$$\Delta\mathbb{E} = \underbrace{\sum_k \left(P_k^{+} - P_k^{-}\right) m_k^{-}}_{\text{extensive: switching}} + \underbrace{\sum_k P_k^{+} \left(m_k^{+} - m_k^{-}\right)}_{\text{intensive: amounts}}. \tag{2}$$

The extensive part is the change in who expects a rise or a fall, valued at the amounts expected before the turn; the intensive part is the change in the amounts, valued at the shares after it.

Figure 6 reports both parts, with 95 percent intervals from a bootstrap that resamples calendar years. In both surveys, most of the jump comes from households switching direction. In the SCE, switching accounts for about 90 percent of the jump.[12] In Michigan, switching accounts for more than the whole jump, and changes in amounts slightly offset it. The step is therefore in *whether* households expect a rise or a fall, not in how much they expect once they do.

Figure 7 shows where this comes from in the SCE. At a perceived change of zero, the share expecting a fall drops and the share expecting a rise jumps, while the size of the expected rise does not change. Away from zero, the amounts do grow with the size of the perceived change, in both directions; this is the response to size, measured by the slopes in (1), and is separate from the step. Michigan shows the same pattern (Appendix Figure D.1): when national growth turns positive, the share answering "up" jumps by 18.4 points, mostly at the expense of "same", and those answering "up" do not give larger numbers. Their numbers even dip slightly, which the band model below explains.

[12] Both the jump decomposed here, $\Delta\mathbb{E}$, and the step J in (1) estimate the jump in the mean at zero from straight-line fits on each side of zero, but they fit different lines. J fits one line to all expectations. $\Delta\mathbb{E}$ fits separate lines to the shares expecting a rise and a fall, and to the amounts among the respondents who expect each, and multiplies them at zero. Because the amounts are fitted only on those who give them, and a product of two lines is not a line, the two estimates differ slightly, by less than the standard error of the step.

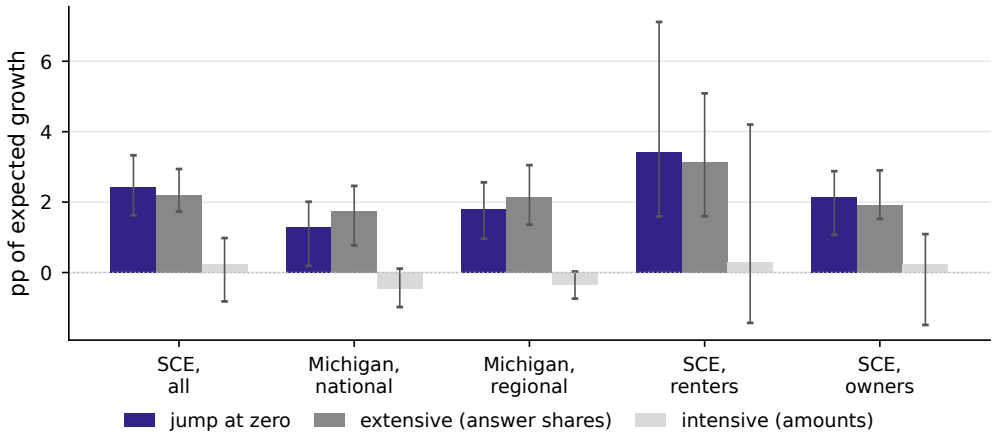
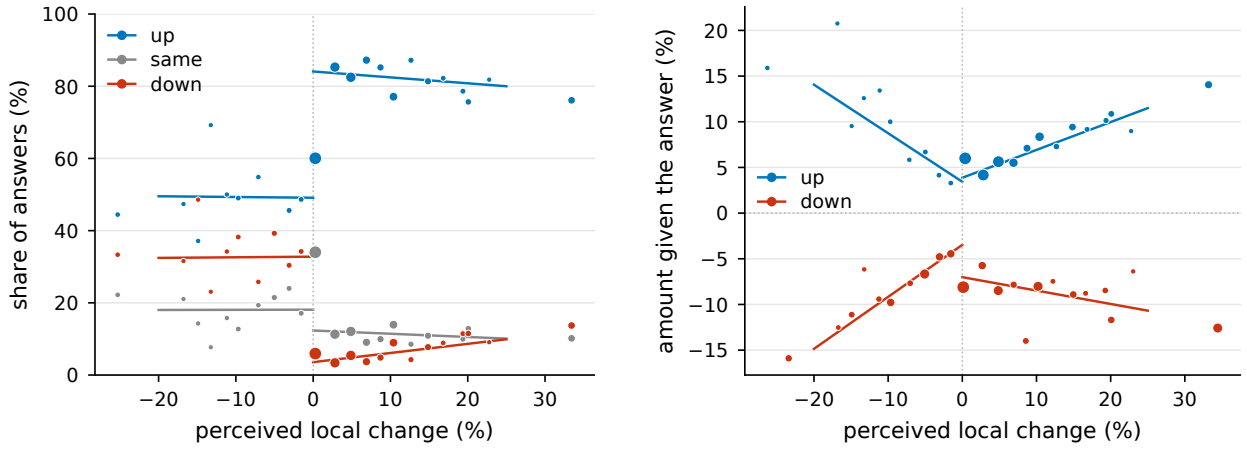


Figure 6: Exact decomposition of the jump in the mean expectation at zero, equation (2): the jump of the fitted mean, the extensive part and the intensive part, with 95% intervals from 999 bootstrap draws resampling calendar years. SCE, perceived local change; Michigan, national and regional growth.



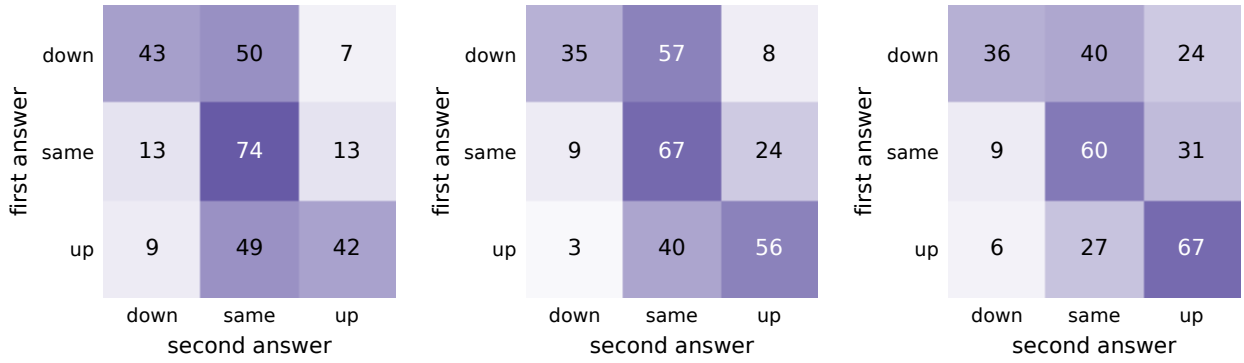
(a) Shares of answers **(b)** Amounts given the answer

Figure 7: Shares and amounts in the SCE, by the perceived past-year local change. (a) Shares expecting a rise, no change and a fall. (b) Average expected change among those expecting a rise and among those expecting a fall. Kinked fits.

3.7.2 Within Households

The SCE has no panel, but the Michigan re-interviews follow the same households six months apart and show both margins within households (Figures 8 and 9). First, households change their answer when the national sign changes, and they do so gradually. Households answering "same" switch to "up" most often when national growth is positive at both interviews, less often when it turns positive between them, and least often when it is negative at both. Households thus follow the new sign with a lag, consistent with sticky information (Mankiw and Reis, 2002; Carroll, 2003). Second, households who give the same answer twice barely revise the amount.

Within households as across them, the sign of past growth moves the answer, while its size moves the number only a little (Appendix D). Because these results follow the national cycle, they carry the same caveat about inflation as the Michigan step: they show how households adjust, not where the step comes from.



(a) Negative at both **(b)** Turned positive **(c)** Positive at both

Figure 8: Answer transitions within households: Michigan interviews six months apart, by the sign of national growth at the two interviews (negative at both, turned positive, positive at both). Each cell is the share of second answers (columns) given the first answer (rows).

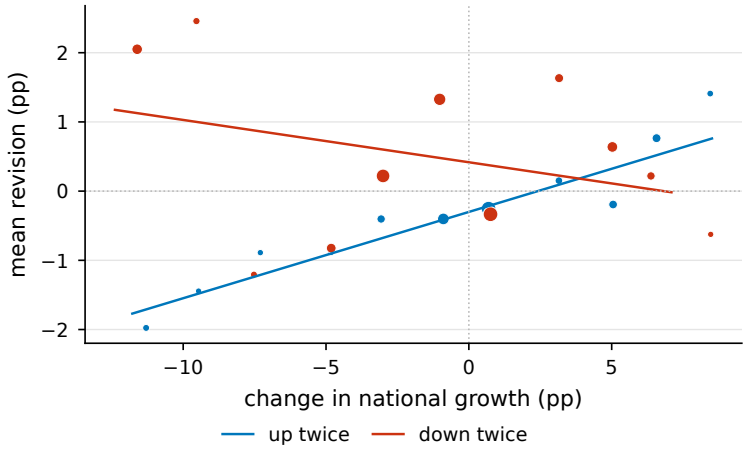


Figure 9: Revisions among unchanged answers: mean revision of the reported number against the change in lagged national growth between the two Michigan interviews, for pairs answering "up" twice and "down" twice.

3.7.3 From Answers to Beliefs

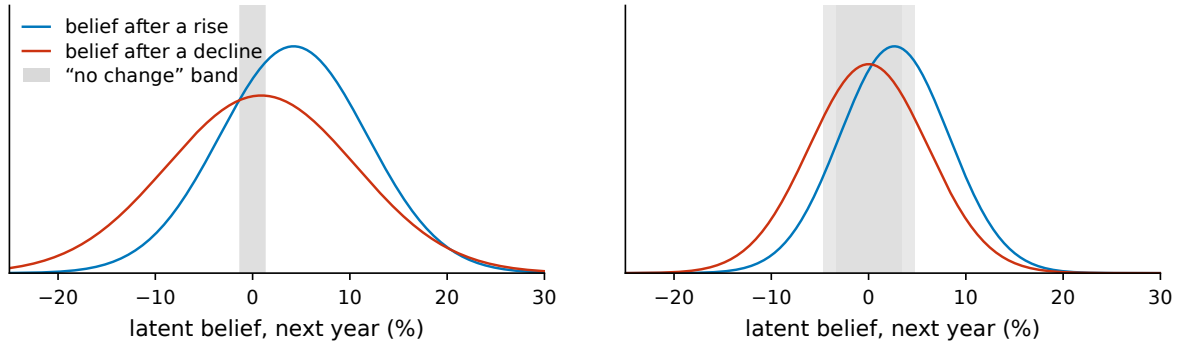
Survey answers are coarse reports of beliefs. To separate the two, I model each answer as the report of a latent belief:

$$b^*_{it} = \tilde{a} + \tilde{J}\,\mathbf{1}[s_t \geq 0] + \tilde{b}^-\,s_t^- + \tilde{b}^+\,s_t^+ + \sigma(s_t)\,\epsilon_{it} \tag{3}$$

The respondent reports as "no change" if her belief lies inside a band, $|b^*_{it}| < c(s_t)$, and otherwise as her belief plus reporting noise. To allow for asymmetry, the band and the dispersion may differ by the sign of past growth. I estimate the model by maximum likelihood (Appendix E). Figure 10 shows how the band turns beliefs into answers, and Figure 11 compares latent beliefs with reported answers. The two surveys define "no change" differently: in Michigan it is an explicit option, while in the SCE it means giving the same dollar value a year ahead as today.

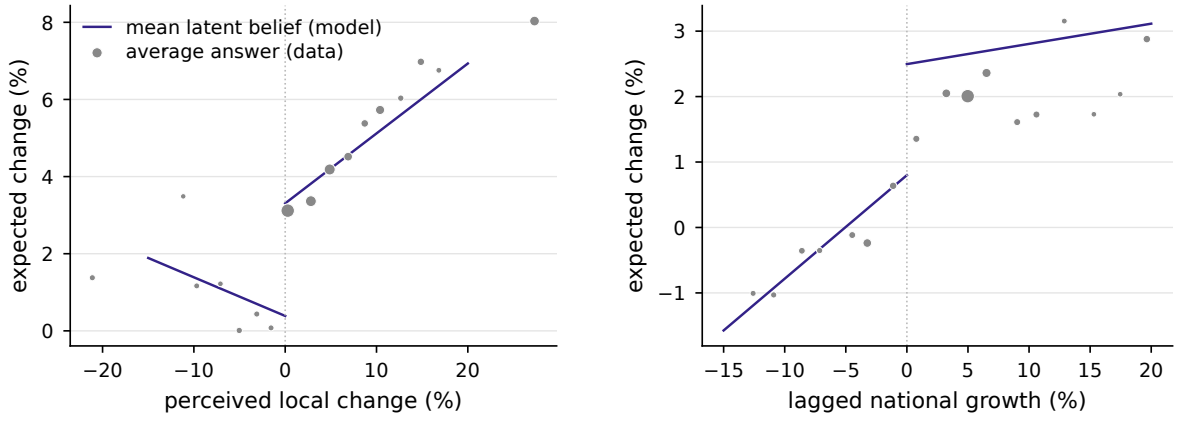
The band model gives two lessons. In the SCE, where "no change" means giving exactly the same value a year ahead, the reports measure beliefs almost exactly: the step is in what households

believe, not in how they report it. In Michigan, "about the same" is a loose answer that households keep giving even as their beliefs move around zero, so answers understate the step in beliefs. Households there also need a stronger belief to report any change after prices fall, a sign that they are more uncertain when prices are falling, in line with evidence that coarse and rounded answers reflect uncertainty (Manski and Molinari, 2010; Binder, 2017). This also explains the small dip in the "up" number at the turn: the households who switch to "up" are those whose beliefs have only just become large enough to report.



(a) SCE **(b)** Michigan, national growth

Figure 10: A model of the answers: the estimated belief distribution after a rise (blue) and after a decline (red) in past prices, with the "no change" band (gray; the band after a decline is wider, the lighter outer shade).



(a) SCE **(b)** Michigan, national growth

Figure 11: Beliefs against answers: the mean latent belief from the band model (line) and the average answer in the data (dots), against past growth.

3.7.4 A Step by Tenure?

The SCE surveys renters as well as owners, while Michigan surveys only owners. Renters' step is about twice owners', both in answers and in latent beliefs (Figure 12), and for both groups it comes from switching direction rather than from amounts. The difference is imprecise, however: with about two thousand renters over ten survey years, it is significant with the year bootstrap of the band model ($p = 0.018$) but not with the wild cluster bootstrap on answers ($p = 0.120$). I therefore treat it as suggestive, in line with evidence that renters and owners form house-price expectations differently (Kindermann et al., 2021). The model gives all households the same belief rule and uses renter-specific rules only as a robustness check.

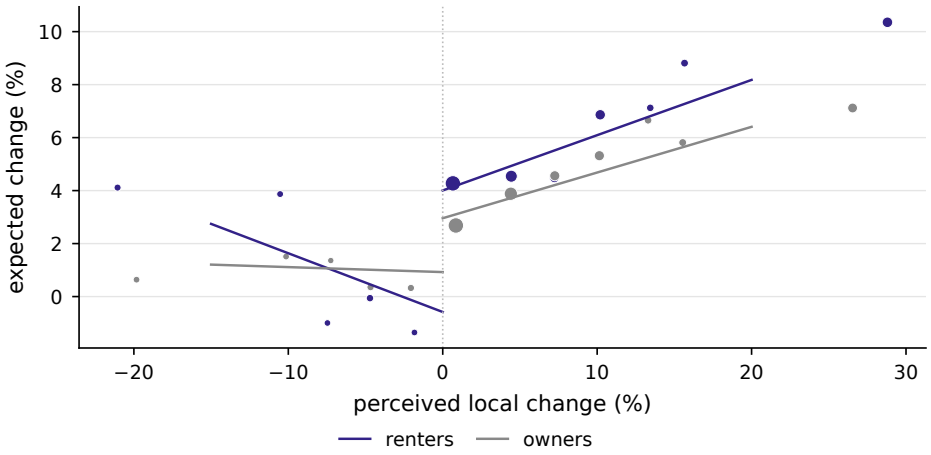


Figure 12: Renters and owners: binned expectations against the perceived past-year local change, with the fit of (1), separately for renters and owners. SCE.

3.8 Interpretation

These facts are consistent with models in which households respond to coarse features of their environment, such as categorical thinking, sparse attention, information costs or a fixed cost of updating (Mullainathan, 2002; Gabaix, 2014; Sims, 2003; Reis, 2006). A fixed cost of updating is particularly attractive here, because a single primitive would generate both the "no change" band and the gradual switching after a turn. These answers should, however, be read with care: only 41–45 percent of respondents hold an expectation for home-price growth before being asked (Andre et al., 2026). I do not study how this affects the results, but two implications seem likely: answers formed on the spot can add to the mass of "same" in Michigan, and households who form an expectation only when asked may rely precisely on salient signals such as the direction of recent prices. Neither, however, can create the step in the SCE, which has no "no change" category.

The measurement of these facts can be improved in several ways. Finer geography, such as ZIP or county identifiers in the SCE and in Michigan, would provide more variation in actual local declines and allow a test of the step at the actual local sign, not only the perceived one. A panel would show how expectations evolve within households: the SCE core survey re-interviews households monthly for up to a year, but asks the housing questions only once a year. Asking the same respondents both Michigan's qualitative question and the SCE's dollar values, together with a probabilistic question, would separate beliefs from how they are reported and help measure uncertainty. Finally, a dedicated survey with an information experiment that varies the direction of past local price changes separately from their size, and that includes enough renters, would allow a causal test of sign dominance.

4 Model

The evidence presented so far concerns house-price expectations. This section asks how sign dominance matters for the decisions households face. To explain the logic in the simplest possible way, I start with a two-period model that delivers the main mechanism and intuition. To better understand the consequences of such belief formation, I then extend it to a partial-equilibrium life-

cycle model with beliefs taken directly from the SCE, and describe its parameterization. Although a general-equilibrium model is not derived here, it is the natural next step to complete this analysis.

In this paper, I do not derive a model to rationalize why households form expectations in this way. Theories in which costs of updating expectations or of acquiring information lead to such belief formation (Sims, 2003; Reis, 2006; Alvarez et al., 2011) are left for future research.

4.1 A Simple Model

Setup. A household lives two periods, $t = 0, 1$. It receives income y_0 and y_1, has utility $\ln c_0 + \beta \ln c_1$, and can save or borrow an amount a at the gross interest rate $R = 1 + r$. A house of size H costs p_0 per unit at $t = 0$ and $(1 + g)p_0$ at $t = 1$. The household does not know g: it believes growth will be $\hat{g}$ and acts as if this were certain. Its welfare is evaluated under the true g, as in Brunnermeier et al. (2014). I consider three households that differ only in what they do with the house. Appendix F contains the full derivations.

An owner who sells. The owner holds the house and sells it at $t = 1$. Her budget constraints are $c_0 = y_0 - a$ and $c_1 = y_1 + Ra + (1 + \hat{g})p_0H$. The Euler equation $1/c_0 = \beta R/c_1$ implies $c_1 = \beta R c_0$, and the lifetime budget then gives

$$c_0 = \frac{1}{1+\beta}\left[y_0 + \frac{y_1 + (1+\hat{g})p_0H}{R}\right], \qquad \frac{\partial c_0}{\partial \hat{g}} = \frac{1}{1+\beta}\,\frac{p_0H}{R} > 0. \tag{4}$$

The response to expected growth is the propensity to consume out of wealth, $1/(1 + \beta)$, times the present value of one point of growth on the house, p_0H/R: the same structure as the housing-wealth effect of Berger et al. (2018), here for an expected rather than a realized gain.

A renter who buys. The renter buys the house at $t = 1$, so $c_1 = y_1 + Ra - (1 + \hat{g})p_0H$. The same steps give $\partial c_0/\partial \hat{g} = -p_0H/[(1+\beta)R]$. The sign is reversed: higher expected prices make the future purchase more expensive, so the renter consumes less and saves more. Higher house prices are a gain only for net sellers (Buiter, 2010). The renter's response is also in line with Chopra et al. (2025).

The cost of a belief error. Evaluated under the true g, the owner's utility as a function of saving is

$$V(a) = \ln(y_0 - a) + \beta \ln\big(y_1 + Ra + (1+g)p_0H\big).$$

Acting on $\hat{g}$ instead of g moves saving from its optimum a^* to $\hat{a}$. Because $V'(a^*) = 0$, the loss is second order in the belief error (Appendix F):

$$L \equiv V(a^*) - V(\hat{a}) \approx \frac{h^2(\hat{g} - g)^2}{2\beta(1+\beta)}, \qquad h \equiv \frac{p_0H}{Rc_0}.$$

As a fraction λ of consumption in both periods,

$$\lambda \approx \frac{L}{1+\beta} = \frac{h^2(\hat{g} - g)^2}{2\beta(1+\beta)^2}. \tag{5}$$

On the saving margin, belief errors move saving a lot but cost little in welfare, as in Akerlof and Yellen (1985); the cost rises with the square of housing exposure h.

A renter who chooses when to buy. The third household starts with liquid wealth A and either buys now, or rents for a year at $\varphi p_0 H$ and buys next year. Each purchase costs a share F of the price. Both options end with the household owning the house; they differ only in liquid wealth at the end of $t = 1$:

$$\text{buy now: } R\big[A - (1+F)p_0 H\big], \qquad \text{wait: } R\big[A - \varphi p_0 H\big] - (1+F)(1+g)p_0 H.$$

The difference is the gain from buying now,

$$\Delta(g) = (1+F)\, p_0 H\, (g - \hat{g}^*), \qquad \hat{g}^* \equiv r - \frac{R\varphi}{1+F}. \tag{6}$$

The household buys now if it expects appreciation above $\hat{g}^*$, the interest rate net of the rent yield.

Aggregation and the wave of purchases. Renters differ in their thresholds $\hat{g}^*$, because rents, transaction costs and tastes differ; let G be the distribution of thresholds. A renter buys if her belief exceeds her threshold, so the share of renters buying is $G(\hat{g})$. If beliefs step at zero, $\hat{g} = \underline{g}$ after a fall in prices and $\underline{g} + J$ after a rise, the share buying jumps when past growth turns positive by

$$G(\underline{g} + J) - G(\underline{g}) \approx G' J, \tag{7}$$

the density of renters whose thresholds lie between the two beliefs, times the step. Under linear beliefs, the share moves smoothly and there is no wave.

The simple model shows that the shape of beliefs matters through the timing of purchases, but it does not say how large the wave is. That depends on G', the density of renters near their threshold when prices turn, which has no analytical solution here and which the life-cycle model below determines.

4.2 Why a Quantitative Model?

The simple model delivers the mechanism, but it takes as given three objects that determine its size. The first is the density G': how many renters are close to buying when prices turn depends on how wealth, income and the down payment evolve over the life cycle. The second is who can act on a belief: a renter who cannot afford the down payment does not buy whatever she expects, so borrowing limits decide which beliefs matter. The third is how beliefs and decisions interact over time: a household that waits one year re-forecasts the next year, and an owner who expects falling prices weighs the transaction cost of selling against the expected loss. A quantitative model therefore needs income risk, a down payment, a life cycle, a costly discrete tenure choice, and beliefs that are revised every year. The model below has these features.

Throughout, the model takes house prices as given. Closing it in general equilibrium, so that the timing of purchases feeds back into prices, is left for the next stage of this project.

4.3 The Life-cycle Model

Households. The economy is populated by households who enter at age 25, retire at 65 and die after age $T = 79$; there is no mortality risk. A household of age j is described by its liquid wealth a, its income shock z, and whether it owned a house last year, o_-. Time is discrete and a period is a year. Each year, events occur in the following order:

1. the household observes its income shock, the current house price q_t and last year's price growth s_t;
2. it forms its forecast of future prices;
3. it draws taste shocks for renting and owning and chooses its tenure for the year;
4. it chooses consumption and saving;
5. it ages one year; after age T it dies and leaves its wealth as a bequest.

Income. While working, log income is

$$\log Y_j = \log \kappa_j + z_j, \qquad z_{j+1} = \rho_z z_j + \varepsilon_{j+1}, \qquad \varepsilon \sim N(0, \sigma_z^2), \tag{8}$$

where κ_j is a deterministic age profile. The persistent shock z is discretized on eleven points (Tauchen, 1986). In retirement, a household receives a fraction ω of the age profile at its last working age, with no income risk. Income risk while working is uninsurable: the only asset is a liquid bond.

Preferences. A household values nondurable consumption C, housing services S and the wealth W it leaves at death,

$$\mathbb{E}\left[\sum_{j=25}^{T} \beta^{j-25}\, u(C_j, S_j) + \beta^{T-24}\, \Psi\, \frac{W^{1-\sigma}}{1-\sigma}\right], \qquad u(C,S) = \frac{\left(C^{\alpha} S^{1-\alpha}\right)^{1-\sigma}}{1-\sigma}. \tag{9}$$

Housing services are $S = H^R$ for a renter and $S = \chi H$ for an owner: a renter lives in a home of size H^R, an owner in a house of size H, and the premium $\chi \geq 1$ stands for the advantages of owning the model leaves out, such as tax treatment and security of tenure. Preferences over consumption and housing services are Cobb–Douglas, as in Berger et al. (2018). The last term is a bequest motive: utility from the wealth left at death.

Housing and the budget constraint. Let $o_t \in \{0, 1\}$ indicate owning in year t, and let cash on hand be $x_t = Ra_t + Y_t$. The budget constraint in year t is

$$C_t + a_{t+1} = x_t - (1 - o_t)\, \varphi q_t H^R - o_t\, \delta q_t H - (o_t - o_{t-1})^{+}(1 + F)\, q_t H + (o_{t-1} - o_t)^{+}(1 - F)\, q_t H. \tag{10}$$

Consumption plus saving equals cash on hand minus four housing flows. A renter pays rent $\varphi q_t H^R$, a fixed share φ of the value of her home. An owner pays maintenance $\delta q_t H$, which offsets depreciation, as in Kaplan et al. (2020). A household that buys pays the price plus a transaction cost, $(1 + F)q_t H$; a household that sells receives the price net of the transaction cost, $(1 - F)q_t H$. An owner's mortgage is part of a_t, so an owner who sells repays it out of the proceeds. Tenure can change in either direction in any year. Wealth left at death is $W = Ra' + o\,(1 - \delta)\, qH$, with a',

o and q in the last year of life: liquid wealth with interest plus the house, net of depreciation, with no transaction cost.

Borrowing. Liquid wealth a is negative for an owner with a mortgage. Renters cannot borrow, $a_{t+1} \geq 0$. A buyer can borrow up to a share θ of the price, $a_{t+1} \geq -\theta q_t H$. An owner who keeps her house can repay her mortgage but not increase it in nominal terms, so that in real terms $a_{t+1} \geq \min(a_t, 0)/(1+\pi)$: her real debt falls with inflation unless she repays more. There is no refinancing, no home-equity extraction and no default. Two implications follow: a fall in prices never forces an owner to repay, and an owner cannot consume an expected capital gain without selling.[13]

Prices. House prices are nominal. With constant inflation π, the real price is $q_t = Q_t/(1+\pi)^t$, where Q_t grows at nominal rate g_t. Because beliefs depend on the sign of nominal growth, the step sits at nominal zero, as in the data. Households take the price path as given.

Beliefs. Households do not know future prices. In year t, a household that has seen past growth s_t expects next year's nominal growth

$$\hat{g}(s_t) = \begin{cases} 0.386\% & \text{if } s_t < 0, \\ 3.306\% & \text{if } s_t \geq 0. \end{cases} \tag{11}$$

The two values are the SCE band model's mean latent belief just below and just above a perceived change of zero (Section 3.7). For later years, the forecast reverts to long-run growth $\bar{g}$ at rate ρ, the persistence of regional house-price growth:

$$g^e_{t+k} = \bar{g} + \big(\hat{g}(s_t) - \bar{g}\big)\rho^{k-1}.$$

Households also know that their forecast is uncertain. Today's price is known, but each future log price differs from the forecast by an independent normal shock, with standard deviation $\sigma_q = 4.6$ percent, the size of one-year forecast errors for national house prices, and mean $-\sigma_q^2/2$, so that the forecast is the expected price. The shock moves everything priced off the house (the sale value, the purchase price and loan limit, rent and maintenance) except the house left as a bequest. No belief parameter is chosen to fit households' decisions.

How beliefs are revised. A household solves its problem as if the forecast, with its uncertainty, were the true price process. It takes this year's decisions, observes next year's price and growth, forms a new forecast and solves again. This is anticipated utility (Cogley and Sargent, 2008): households do not anticipate that they will revise their beliefs.[14]

[13] These assumptions make housing illiquid, as in Kaplan et al. (2020), but without their home-equity line or a choice of house size. Therefore, owners' spending barely responds to expected prices in the model (Section 4.4).

[14] A household with internally consistent subjective beliefs would take its future revisions into account (Adam and Marcet, 2011). I use anticipated utility as an approximation here to keep the problem tractable.

The household problem. Given its forecast, a household maximizes (9) subject to the budget constraint (10) and the borrowing limits. Recursively, with state (a, z, o_-) at age j,

$$\begin{aligned} V_j(a, z, o_-) &= \mathbb{E}_\epsilon \Big[\max_{o \in \{0,1\}} \; v_j(a, z, o_-, o) + \nu \, \epsilon_o \Big], \\ v_j(a, z, o_-, o) &= \max_{C,\, a'} \; u\big(C, S(o)\big) + \beta \, \mathbb{E}\big[V_{j+1}(a', z', o)\big], \end{aligned} \tag{12}$$

where $v_j(\cdot, o)$ is the value of choosing tenure o and the expectation is over next year's income shock and price shock; V_{T+1} is the bequest term. The taste shocks ϵ_o are extreme-value with scale ν, so a household chooses to own with probability

$$P_j(a, z, o_-) = \frac{\exp\big(v_j(a, z, o_-, 1)/\nu\big)}{\exp\big(v_j(a, z, o_-, 0)/\nu\big) + \exp\big(v_j(a, z, o_-, 1)/\nu\big)}. \tag{13}$$

Without taste shocks, households with similar wealth and income would all switch tenure at the same age; with them, ownership rises smoothly with age. The solution consists of the ownership probability (13) and, for each tenure, a consumption rule $C_j(a, z, o_-, o)$ and a saving rule $a'_j(a, z, o_-, o)$. Together with the income process and the price path, these rules move the distribution of households over (j, a, z, o_-) from one year to the next. The problem is solved by the discrete-continuous endogenous grid method (Iskhakov et al., 2017); Appendix G gives the details.

Two properties of this problem are worth emphasizing. First, beliefs enter only through the forecast price, which affects the cost of renting, the cost of buying later, the loan limit and the value of the house at sale or death. The model's main departure from a standard life-cycle housing model is therefore the belief rule (11). Second, the problem is non-convex: the tenure choice is discrete and switching tenure is costly, so first-order conditions are not sufficient. The taste shocks smooth the choice probabilities, which helps make the model more realistic.

Two households. I compare two households that differ only in their beliefs. The *step household* forms beliefs with rule (11) and solves (12) every year. The *full-information household* has the same preferences, income, constraints, calibration and random draws, but knows the actual future price path and faces no price risk. The comparison is the life-cycle version of the simple model: both act on their beliefs, and outcomes are evaluated under the true path. Full information is therefore an upper bound, because it includes the value of timing the market perfectly.

4.4 Parameterization

Strategy. The calibration has three goals. The beliefs must come from the data, so that no belief parameter is tuned to fit decisions. The economy must have the right number of owners and renters, at every age. And households must respond to transfers and to expected prices as in the data. I calibrate in the steady state, in which prices grow at long-run growth $\bar{g}$ and households hold the SCE rule. Table 4 lists every parameter.

Parameters set externally. Preferences, the interest rate, the income process, the rent-to-price ratio and maintenance follow Berger et al. (2018). Buyers can borrow 0.900 of the price, a 10 percent down payment as for first-time buyers (National Association of Realtors, 2024; Kaplan

et al., 2020). The transaction cost, the bequest weight, the replacement rate and the two house sizes are set by assumption: the bequest weight is large enough for the motive to bind, and the house is worth about four times mean income at age 25. Inflation and long-run growth are the sample means of CPI inflation and Case-Shiller growth. The persistence ρ comes from regional house-price growth, the price risk σ_q from the national forecasting regression described above, and the belief rule from the SCE band model. The taste-shock scale ν is set so that ownership rises smoothly with age: it is the smallest value, under the first calibration of the model, at which purchases do not bunch at a single age, and it is kept at that value.

Parameters calibrated internally. The ownership premium χ, the discount factor β and mean initial wealth w_0 are chosen jointly to match homeownership by age in the Survey of Consumer Finances around 1998 (Board of Governors of the Federal Reserve System, 2001) and an annual marginal propensity to consume of about one half (Fagereng et al., 2021).[15] Table 5 shows the fit: the model matches ownership closely at every age, while its marginal propensity to consume, 0.4, is somewhat below the target. Households enter with little wealth: on average 0.2 times mean income at 25.

Untargeted behavior. Homeownership by single year of age is not targeted; within each age group, it rises smoothly with age (Figure 13). For a one-point rise in expected average growth over ten years, renters cut spending by 4.67 percent, close to the 3.6 percent in the experiment of Chopra et al. (2025), and owners barely respond, as in that experiment. Owners respond little because, as explained above, an owner cannot consume an expected gain without selling.

Table 4: Parameters

	Parameter	Value	Source
Beliefs	After a fall / a rise (%)	0.386 / 3.306	SCE band model
	Persistence of forecasts ρ	0.644	persistence of regional growth
	Price risk σ_q (%)	4.56	national forecast errors
Demography	Entry; retirement; last age	25; 65; 79	assumption
Preferences	Curvature σ	2.0	Berger et al. (2018)
	Weight on consumption α	0.8875	Berger et al. (2018)
	Bequest weight Ψ	500	assumption
	Taste-shock scale ν	0.040	smooth ownership by age
Income	Persistence ρ_z; s.d. σ_z	0.91; 0.21	Berger et al. (2018)
	Replacement rate ω	0.50	assumption
Housing	House sizes H; H^R	3.0; 1.5	assumption
	Rent-to-price ratio φ	0.06	Berger et al. (2018)
	Maintenance δ	0.022	Berger et al. (2018)
	Transaction cost F	0.05	assumption
	Loan-to-value θ	0.900	National Association of Realtors (2024)
Prices	Inflation π; growth $\bar{g}$ (%)	2.78; 4.37	CPI; Case-Shiller
	Gross interest rate R	1.024	Berger et al. (2018)
Calibrated	Ownership premium χ	3.271	Table 5
	Discount factor β	0.900	Table 5
	Mean initial wealth w_0	0.2111	Table 5

[15] Homeownership is targeted in five age groups, from 25–34 to 65–79, in the 1995, 1998 and 2001 waves. The marginal propensity to consume is the same-year change in consumption after an unexpected transfer of 1 percent of mean working-age income, averaged over working ages 26 to 64. The three parameters minimize the sum of squared gaps to these six moments, scaled by 2 percentage points for ownership and 0.05 for the marginal propensity to consume.

Table 5: Calibrated parameters: moments in the data and in the model

Moment	Data	Model
Targeted		
Homeownership, ages 25–34 (%)	45.8	45.8
Homeownership, ages 35–44 (%)	66.5	67.1
Homeownership, ages 45–54 (%)	75.3	74.2
Homeownership, ages 55–64 (%)	81.8	81.2
Homeownership, ages 65–79 (%)	80.7	81.9
Annual marginal propensity to consume	0.500	0.414
Implied, not targeted		
Homeownership, ages 25–79 (%)	69.0	69.2
Calibrated values: $\chi = 3.271$, $\beta = 0.900$, $w_0 = 0.2111$		

Ownership: SCF 1995, 1998 and 2001, household heads of each age; ages 25–79 weighted by the SCF age distribution of heads. Marginal propensity to consume: Fagereng et al. (2021).

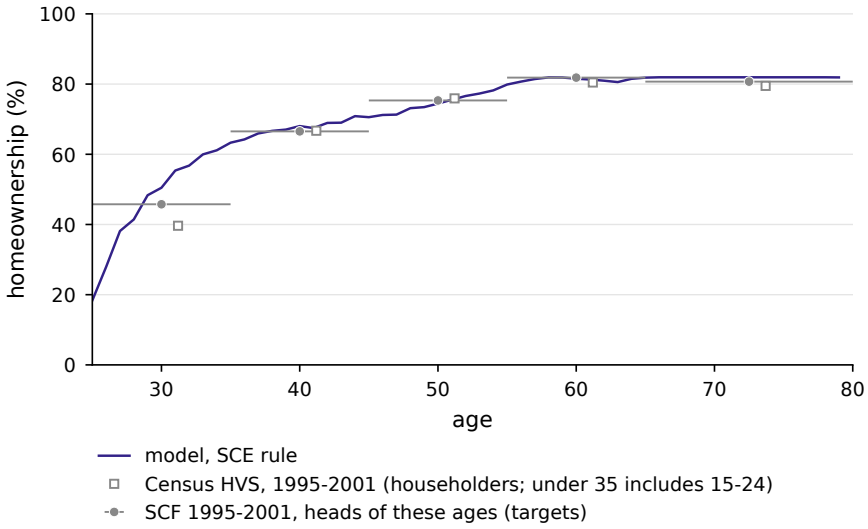


Figure 13: Homeownership by age in the steady state: the model by single year of age (line), against the SCF 1995–2001 rates for household heads in the five targeted age groups (dots; bars span each group) and the Census HVS rates for householders over the same years (squares; under 35 includes ages 15–24).

5 Results

This section presents the quantitative results. Throughout, I compare two households that differ only in their beliefs: the *step household*, which forecasts prices with the SCE rule, and the *full-information household*, which knows the actual price path. I first describe the experiment. I then show when the two households buy, check the simple model's prediction for the size of the wave, describe the households who mistime their purchase, and examine how the results depend on the step and on frictions. I end with welfare and with what the model leaves out.

The experiment. The economy starts in the 1998 steady state of the calibrated model, and house prices then follow actual Case-Shiller growth through 2025. From 1999, a new cohort of renters enters at age 25 each year, with income drawn from its stationary distribution and initial wealth $w_0 e^{\xi}$, with $\xi \sim N(-\frac{1}{2}, 1)$, so that mean initial wealth is w_0. The two households start from the same

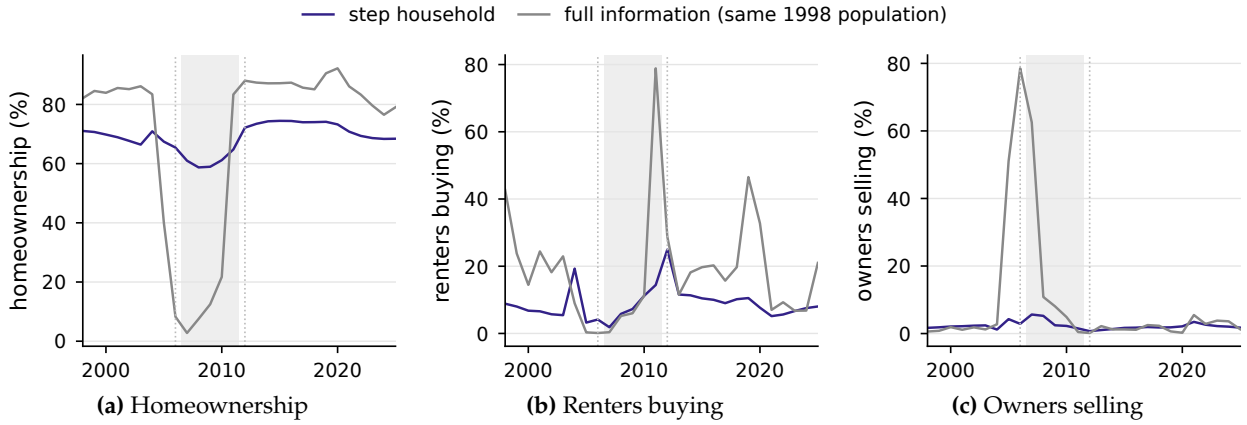
population and receive the same income and taste shocks; only their beliefs differ. The welfare loss of the step belief is the consumption equivalent k: the proportional cut in the full-information household's consumption, housing services and bequest in every remaining year that makes its lifetime utility equal to the step household's, both evaluated under the actual price path.[16]

5.1 The Timing of Purchases

Figure 14 shows homeownership, the share of renters who buy and the share of owners who sell, for both households. The step household buys most when its belief turns: 24.9 percent of its renters buy in 2012, the first year in which past growth is positive again. The full-information household buys at the bottom of the market, a year earlier: 78.9 percent of its renters buy in 2011.

The intuition follows directly from the simple model. Throughout the bust, the step household expects only 0.386 percent nominal growth, below the interest rate net of the rent yield for most renters, so it keeps renting. When past growth turns positive, its expected growth jumps to 3.306 percent, and every renter whose threshold lies between the two beliefs crosses it in the same year. The full-information household knows that prices are about to rise, so its renters buy before the turn, when prices are lowest. The step household is therefore late by exactly the time it takes for past growth to change sign.

Two further features of Figure 14 deserve comment. Step owners do not sell into the bust: their expected growth falls, but not by enough to justify paying the transaction cost twice, selling now and buying back later. The full-information household, by contrast, sells before the peak and buys back at the bottom. This is perfect market timing, which no household could achieve, so full information is an upper bound rather than a realistic benchmark. The model is a simplified description of behavior, so the size of these movements should be read with caution. Its clearest prediction is about timing: purchases by renters bunch in the year past growth turns positive, which data on first-time buyers around turning points could test in future work.



(a) Homeownership **(b)** Renters buying **(c)** Owners selling

Figure 14: The step household (SCE rule) against the full-information household: homeownership, renters buying and owners selling, by year. Shaded: the bust after the 2006 peak.

[16] Utility from taste shocks is not scaled, and households that reach an infeasible state under either belief are excluded.

5.2 The Wave of Purchases Is $G'J$

The simple model predicts that the wave of purchases at the turn equals the density of renters near their threshold times the step, (7). The life-cycle model allows a direct check. For each renter in 2012, I compute her buying threshold: the expected growth above which she prefers to buy rather than rent. Figure 15 plots the share of renters whose threshold lies below each value, that is, the share who would buy at that belief. Renters with thresholds in the shaded band, between the belief after a fall and the belief after a rise, do not buy while prices fall but buy once the belief steps up. They make up 17.8 percent of renters, which is the wave the simple model predicts. Holding everything else fixed, the step raises the share of renters buying in 2012 by 15.8 percentage points, close to this prediction.

The wave in the life-cycle model therefore comes from the mechanism of Section 4.1, not from some other feature of the model. This also gives a simple way to think about other settings: the wave is large when many renters are close to buying when prices turn, and when the step is large.

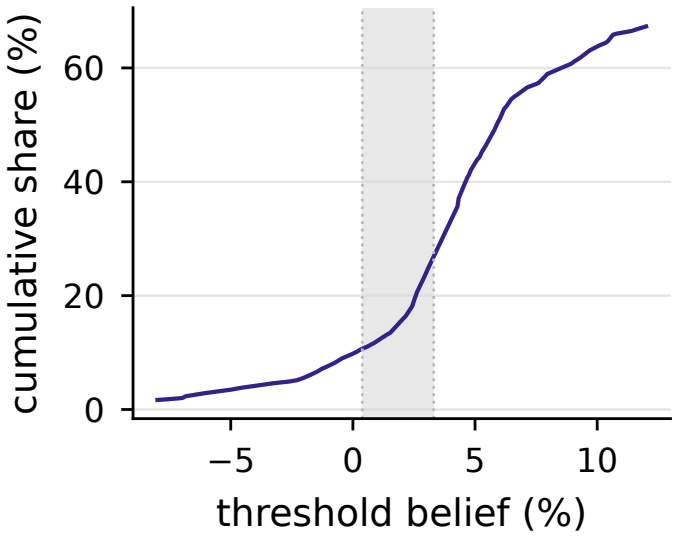


Figure 15: Buying thresholds in 2012: share of renters whose threshold belief lies below each value. The shaded band lies between the beliefs after a fall and after a rise.

5.3 Who Mistimes Their Purchase

Which households does the step affect? I call *mistimers* the renters who would buy in 2011 under full information but buy in 2012 under the step belief; they make up 17.9 percent of renters in 2011. Figure 16 shows why they wait: in 2011, the gain from buying is mostly negative under the step belief, but positive for many renters under the true path. Mistimers are in their late forties on average (Appendix Table G.1) and could have bought in 2011, since a household with the same wealth and income but full information does: it is their belief, not their finances, that holds them back. The poorest renters, by contrast, cannot meet the down payment in either year, so their belief does not affect whether they buy. Expectations therefore matter most for households who are close to buying anyway. These are also the households whose spending responds most to expected prices, because they are saving for a purchase, which is the margin Chopra et al. (2025) document experimentally.

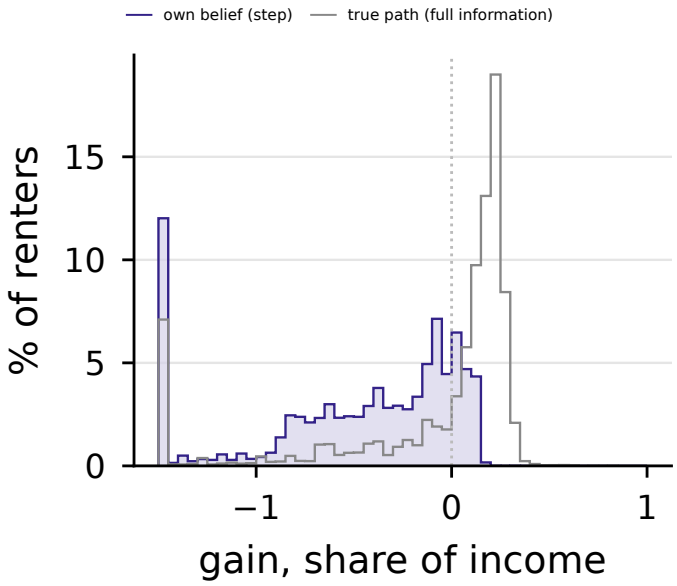
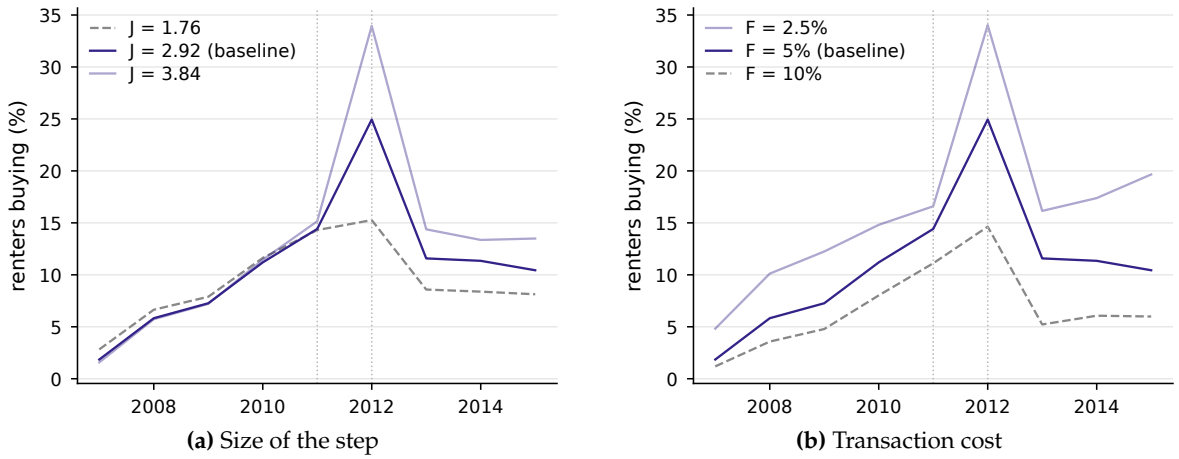


Figure 16: Gain from buying in 2011 rather than renting, as a share of income, for each renter under the step belief and under the true path. The spike at the left edge pools renters for whom buying is not feasible.

5.4 Comparative Statics

Figure 17 varies the size of the step and the transaction cost, keeping the calibration fixed. The size of the wave moves as (7) predicts: it rises with the step and falls with the transaction cost, which spreads thresholds out. With a smaller step, at the low end of its confidence interval, the wave of purchases largely disappears: the step must be large enough to move many renters across their threshold at the same time. The timing, by contrast, does not change: whenever there is a wave, it comes at the turn for the step household and at the trough for the full-information household.[17] The model's main prediction is thus tied to the size of the step measured in the data.



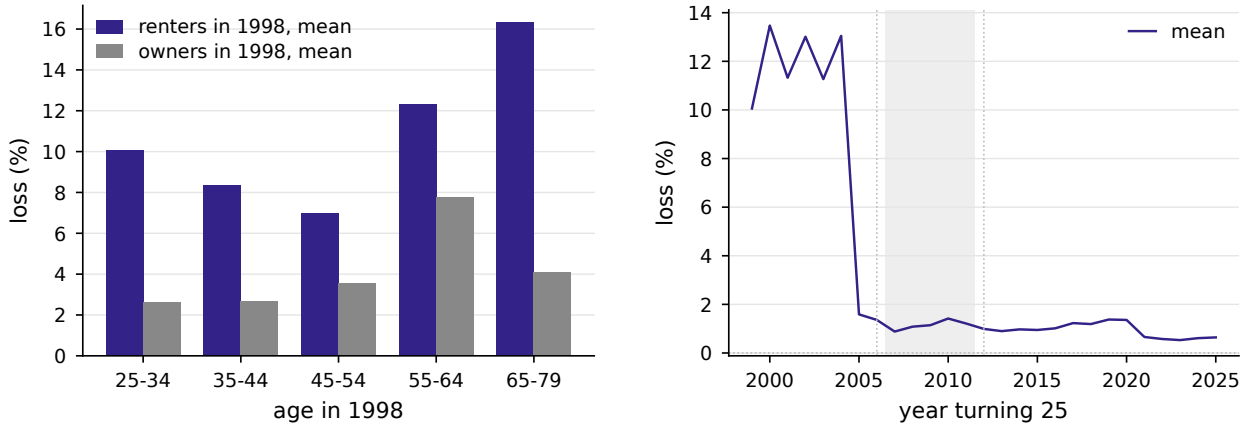
(a) Size of the step **(b)** Transaction cost

Figure 17: Comparative statics, calibration fixed: share of renters buying each year under the step belief, in the baseline and with (a) the step at the low and high ends of its confidence interval, (b) the transaction cost halved (2.5%) or doubled (10%). Dotted lines: the trough (2011) and the turn (2012).

[17] A loan-to-value limit of 0.80 shrinks the wave somewhat, while a limit of 0.95 and other values of the taste-shock scale leave it little changed; with renter-specific belief rules, whose step is larger, the wave is larger and comes in the same year; Appendix Table G.1 reports every version.

5.5 Welfare

Against full information, the step household loses 5.3 percent of lifetime consumption on average, with a median of 2.6 percent (Figure 18). This comparison gives an upper bound on the cost of the step belief, because the full-information household times the market perfectly: it buys early in the boom, sells before the bust, buys back at the bottom and anticipates the rise of 2020–22. The loss does not come from the bequest motive, the ownership premium or price risk, since removing each changes it little (Appendix Table G.1). It is concentrated on households who rent during the boom: renters in 1998 lose more than owners at every age, and cohorts entering before 2005 lose more than ten times as much as those entering after the bust. A tighter measure of the cost of the step would compare the step household with one that knows the price process but not the realized path, across many simulated paths; this is left for the next stage of the project.



(a) By age and tenure in 1998

(b) Entering cohorts

Figure 18: Welfare loss of the step household against full information, as a percentage of lifetime consumption: (a) by age and tenure in 1998; (b) for the cohorts entering each year.

5.6 What the Model Leaves Out

Three simplifications deserve mention. First, rent is a fixed share of the house price. When rent instead follows the CPI rent index, the timing at the turn is unchanged, but step households, who never expect prices to outpace rents, own too little during the boom (Appendix G). Second, there is no default, so a few households on extreme price paths reach infeasible states. Third, owners have one house size and cannot extract equity, which is why their spending barely responds to expected prices. None of these should affect the timing result, which depends only on the step and on how many renters are close to their threshold.

6 Conclusion

This paper shows that households' house-price expectations depend on the direction of the price changes they perceive, beyond their size. In the SCE, households who perceive a rise expect markedly higher growth than households who perceive a fall in the same year, and the same step appears in their expectations for national prices. The step comes mostly from households switching the direction of their forecast rather than revising its size. Embedded in a partial-equilibrium life-cycle model, this step in beliefs may become a step in behavior: renters wait for

prices to turn before buying, a year after the trough.

These findings also have implications for policy. They suggest that measures of household expectations could be improved by tracking the direction of expected changes alongside their size. They may also matter for how monetary policy is transmitted through housing. Finally, they suggest that more accurate, transparent information about local house prices could help align perceptions with actual prices.

This paper leaves for future work analysis of more granular data, with finer geography and repeated interviews of the same households to sharpen the measurement. A dedicated survey with an information experiment could help establish the step causally. Closing the model in general equilibrium would show how the step in beliefs feeds back into prices and whether it amplifies turning points.

References

Adam, K. and A. Marcet (2011). Internal rationality, imperfect market knowledge and asset prices. *Journal of Economic Theory 146*(3), 1224–1252.

Adam, K., O. Pfäuti, and T. Reinelt (2025). Subjective housing price expectations, falling natural rates, and the optimal inflation target. *Journal of Monetary Economics 149*.

Akerlof, G. A. and J. L. Yellen (1985). Can small deviations from rationality make significant differences to economic equilibria? *American Economic Review 75*(4), 708–720.

Aladangady, A., J. Bricker, A. C. Chang, S. Goodman, J. Krimmel, K. B. Moore, S. Reber, A. H. Volz, and R. A. Windle (2023, October). Changes in U.S. family finances from 2019 to 2022: Evidence from the survey of consumer finances. Technical report, Board of Governors of the Federal Reserve System.

Alvarez, F. E., F. Lippi, and L. Paciello (2011). Optimal price setting with observation and menu costs. *Quarterly Journal of Economics 126*(4), 1909–1960.

Andre, P., F. Chopra, L. Michels, and J. Wohlfart (2026). Do people have economic expectations? CESifo Working Paper 12603, CESifo.

Armantier, O., G. Topa, W. van der Klaauw, and B. Zafar (2017). An overview of the survey of consumer expectations. *Federal Reserve Bank of New York Economic Policy Review 23*(2), 51–72.

Armona, L., A. Fuster, and B. Zafar (2019). Home price expectations and behaviour: Evidence from a randomized information experiment. *Review of Economic Studies 86*(4), 1371–1410.

Bailey, M., R. Cao, T. Kuchler, and J. Stroebel (2018). The economic effects of social networks: Evidence from the housing market. *Journal of Political Economy 126*(6), 2224–2276.

Bailey, M., E. Dávila, T. Kuchler, and J. Stroebel (2019). House price beliefs and mortgage leverage choice. *Review of Economic Studies 86*(6), 2403–2452.

Berger, D., V. Guerrieri, G. Lorenzoni, and J. Vavra (2018). House prices and consumer spending. *Review of Economic Studies 85*(3), 1502–1542.

Binder, C. C. (2017). Measuring uncertainty based on rounding: New method and application to inflation expectations. *Journal of Monetary Economics 90*, 1–12.

Board of Governors of the Federal Reserve System (2001). Survey of consumer finances, 1995, 1998 and 2001. Summary extract public data.

Bottan, N. L. and R. Perez-Truglia (2020). Betting on the house: Subjective expectations and market choices. NBER Working Paper 27412, National Bureau of Economic Research.

Brunnermeier, M. K., A. Simsek, and W. Xiong (2014). A welfare criterion for models with distorted beliefs. *Quarterly Journal of Economics 129*(4), 1753–1797.

Buiter, W. H. (2010). Housing wealth isn't wealth. *Economics: The Open-Access, Open-Assessment E-Journal 4*(2010-22), 1–29.

Burnside, C., M. Eichenbaum, and S. Rebelo (2016). Understanding booms and busts in housing markets. *Journal of Political Economy 124*(4), 1088–1147.

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics 90*(3), 414–427.

Carlson, J. A. and M. Parkin (1975). Inflation expectations. *Economica 42*(166), 123–138.

Carroll, C. D. (2003). Macroeconomic expectations of households and professional forecasters. *Quarterly Journal of Economics 118*(1), 269–298.

Case, K. E., R. J. Shiller, and A. K. Thompson (2012). What have they been thinking? homebuyer behavior in hot and cold markets. *Brookings Papers on Economic Activity*. Fall issue.

Chodorow-Reich, G., A. M. Guren, and T. J. McQuade (2024). The 2000s housing cycle with 2020 hindsight: A neo-kindlebergerian view. *Review of Economic Studies 91*(2).

Chopra, F., C. Roth, and J. Wohlfart (2025). Home price expectations and spending: Evidence from a field experiment. *American Economic Review 115*(7), 2267–2305.

Cogley, T. and T. J. Sargent (2008). Anticipated utility and rational expectations as approximations of Bayesian decision making. *International Economic Review 49*(1), 185–221.

Davezies, L. and T. Le Barbanchon (2017). Regression discontinuity design with continuous measurement error in the running variable. *Journal of Econometrics 200*(2).

De Stefani, A. (2021). House price history, biased expectations, and credit cycles: The role of housing investors. *Real Estate Economics 49*(4).

Fagereng, A., M. B. Holm, and G. J. Natvik (2021). MPC heterogeneity and household balance sheets. *American Economic Journal: Macroeconomics 13*(4), 1–54.

Foote, C. L., K. S. Gerardi, and P. S. Willen (2012). Why did so many people make so many ex post bad decisions? the causes of the foreclosure crisis. NBER Working Paper 18082, National Bureau of Economic Research.

Gabaix, X. (2014). A sparsity-based model of bounded rationality. *Quarterly Journal of Economics 129*(4), 1661–1710.

Gennaioli, N. and A. Shleifer (2018). *A Crisis of Beliefs: Investor Psychology and Financial Fragility*. Princeton University Press.

Glaeser, E. L., J. Gyourko, E. Morales, and C. G. Nathanson (2014). Housing dynamics: An urban approach. *Journal of Urban Economics 81*, 45–56.

Glaeser, E. L. and C. G. Nathanson (2017). An extrapolative model of house price dynamics. *Journal of Financial Economics 126*(1), 147–170.

Gorodnichenko, Y. and D. Sergeyev (2021). Zero lower bound on inflation expectations. NBER Working Paper 29496, National Bureau of Economic Research.

Hansen, B. E. (2000). Sample splitting and threshold estimation. *Econometrica 68*(3), 575–603.

Himmelberg, C., C. Mayer, and T. Sinai (2005). Assessing high house prices: Bubbles, fundamentals and misperceptions. *Journal of Economic Perspectives 19*(4), 67–92.

Iskhakov, F., T. H. Jørgensen, J. Rust, and B. Schjerning (2017). The endogenous grid method for discrete-continuous dynamic choice models with (or without) taste shocks. *Quantitative Economics 8*(2), 317–365.

Jung, J., J. H. Kim, F. Matějka, and C. A. Sims (2019). Discrete actions in information-constrained decision problems. *Review of Economic Studies 86*(6), 2643–2667.

Kaplan, G., K. Mitman, and G. L. Violante (2020). The housing boom and bust: Model meets evidence. *Journal of Political Economy 128*(9), 3285–3345.

Kiesl-Reiter, S., M. Lührmann, J. Shaw, and J. Winter (2025). The formation of subjective house price expectations: The role of perceptions and local economic conditions. IFS Working Paper 25/58, Institute for Fiscal Studies.

Kindermann, F., J. Le Blanc, M. Piazzesi, and M. Schneider (2021). Learning about housing cost: Survey evidence from the German house price boom. NBER Working Paper 28895, National Bureau of Economic Research.

Kuchler, T., M. Piazzesi, and J. Stroebel (2023). Housing market expectations. In R. Bachmann, G. Topa, and W. van der Klaauw (Eds.), *Handbook of Economic Expectations*. Academic Press.

Kuchler, T. and B. Zafar (2019). Personal experiences and expectations about aggregate outcomes. *Journal of Finance 74*(5), 2491–2542.

Landvoigt, T. (2017). Housing demand during the boom: The role of expectations and credit constraints. *Review of Financial Studies 30*(6), 1865–1902.

Landvoigt, T., M. Piazzesi, and M. Schneider (2015). The housing market(s) of San Diego. *American Economic Review 105*(4), 1371–1407.

Ludwig, A., J. Mankart, J. Quintana, and M. Wiederholt (2026). Heterogeneity in expectations and house price dynamics. Working paper, April 2026; earlier version CEPR Discussion Paper 19498 (2024).

Ma, C. (2020). Momentum and reversion to fundamentals: Are they captured by subjective expectations of house prices? *Journal of Housing Economics 49*.

MacKinnon, J. G. and M. D. Webb (2017). Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics 32*(2), 233–254.

Mankiw, N. G. and R. Reis (2002). Sticky information versus sticky prices: A proposal to replace the new keynesian phillips curve. *Quarterly Journal of Economics 117*(4), 1295–1328.

Manski, C. F. and F. Molinari (2010). Rounding probabilistic expectations in surveys. *Journal of Business & Economic Statistics 28*(2), 219–231.

Matějka, F. and A. McKay (2015). Rational inattention to discrete choices: A new foundation for the multinomial logit model. *American Economic Review 105*(1), 272–298.

Mullainathan, S. (2002). Thinking through categories. Working paper, Massachusetts Institute of Technology.

National Association of Realtors (2024). 2024 profile of home buyers and sellers. National Association of Realtors.

Pei, Z. and Y. Shen (2017). The devil is in the tails: Regression discontinuity design with measurement error in the assignment variable. In *Regression Discontinuity Designs: Theory and Applications*, Volume 38 of *Advances in Econometrics*. Emerald.

Poterba, J. M. (1984). Tax subsidies to owner-occupied housing: An asset-market approach. *Quarterly Journal of Economics 99*(4), 729–752.

Reis, R. (2006). Inattentive consumers. *Journal of Monetary Economics 53*(8), 1761–1800.

Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics 50*(3), 665–690.

Survey Research Center, University of Michigan (2025). Surveys of consumers. Institute for Social Research, University of Michigan.

Tauchen, G. (1986). Finite state Markov-chain approximations to univariate and vector autoregressions. *Economics Letters 20*(2), 177–181.

Wang, T. (2023). Perceived versus calibrated income risks in heterogeneous-agent consumption models. Staff Working Paper 2023-59, Bank of Canada.

Webb, M. D. (2023). Reworking wild bootstrap-based inference for clustered errors. *Canadian Journal of Economics 56*(3).

A Data Construction

SCE. The housing module is fielded each February. In 2014–20, respondents give the value of a typical home in their ZIP code today, a year ago and a year from now; the perceived and expected changes are computed from these values. From 2022 the past change is asked directly. The public module carries no geography, so I link respondents by identifier to the core survey files, which provide the state and demographics, and match each housing interview to the core interview in the same month. The national home-price expectation is taken from the core survey's density forecast for the same February.

Michigan. All zeros in the house-price expectation are answers of "stay about the same"; no up or down answer reports zero. "Don't know" answers are excluded. Respondents who give a direction but no percentage enter the analyses of answer shares and the band model through their category.

Regional growth. Averaging Zillow price levels across the metro areas available each month mixes price changes with changes in coverage, which grew sharply in 2008–12. The series used here computes twelve-month growth within each metro observed at both dates and averages across metros in each Census region. State growth averages county growth within each state.

B Inference

Table B.1 reports the main estimates, and the regressions of perceived on actual past price changes behind Section 2.4, under conventional and alternative standard errors. In Michigan, twelve-month outcomes observed monthly overlap, so neighboring months are correlated and month clusters understate uncertainty; in the SCE there are only ten years. In simulations calibrated to these data with few clusters, the wild cluster bootstrap with Webb weights rejects a true null at the 5 percent level in 11.5 percent of samples, against 17.5 percent for the cluster-robust test: both over-reject, but the bootstrap much less. In every table of this appendix, unless its notes say otherwise, standard errors are clustered by state in the SCE and by calendar year in Michigan, and p-values and stars come from a wild cluster bootstrap by survey year.

Table B.1: Inference under alternative standard errors

	Estimate	SE (1)	SE (2)	SE (3)	SE (4)	Bootstrap p
Panel A. SCE. SE: (1) heteroskedasticity-robust; (2) state clusters; (3) year clusters; (4) state and year clusters						
Step J, all respondents	2.815***	0.592 (0.000)	0.506 (0.000)	0.565 (0.000)	0.489 (0.000)	0.005
Step at exactly zero J_0, all respondents	2.945***	0.617 (0.000)	0.547 (0.000)	0.690 (0.000)	0.617 (0.000)	0.001
Slope b^+, all respondents	0.186***	0.019 (0.000)	0.013 (0.000)	0.026 (0.000)	0.023 (0.000)	<0.001
Slope b^-, all respondents	-0.102	0.082 (0.213)	0.076 (0.180)	0.127 (0.419)	0.124 (0.409)	0.458
Step J, owners	2.033**	0.594 (0.001)	0.594 (0.001)	0.538 (0.000)	0.567 (0.000)	0.043
Step J, renters	4.588***	1.521 (0.003)	1.868 (0.014)	1.407 (0.001)	1.632 (0.005)	0.002
Perception slope, year effects (p: slope = 1)	0.395	0.038 (0.000)	0.054 (0.000)	0.057 (0.000)	0.067 (0.000)	0.001
Panel B. Michigan. SE: (1) month clusters; (2) year clusters; (3) two-year blocks						
Step J, national growth	1.284***	0.206 (0.000)	0.373 (0.001)	0.421 (0.002)		0.006
Step J, regional growth	1.766**	0.173 (0.000)	0.413 (0.000)	0.434 (0.000)		0.015
Perception slope (p: slope = 1)	0.628	0.043 (0.000)	0.068 (0.000)	0.063 (0.000)		0.005

Notes: Normal p-values in parentheses. Bootstrap p: all 6^{10} Webb sign draws enumerated (SCE); 1,999 draws (Michigan). Perception rows: perceived on actual past change (SCE: state growth, N = 8,028; Michigan: change in the reported home value on regional growth, N = 28,997); p-values test a slope of one.

C Robustness of the Step

Table C.1 collects the checks of Section 3.5 for the SCE, in the order in which the text discusses them. Panel A reports the step J in (1) with demographic controls, state and state-by-year fixed

effects, and mood controls; without mechanically symmetric answers and by question format; without round answers; and with the national expectation as the outcome (Section 3.6). Panel B reports the mood placebo: the step in respondents' other expectations. Panel C reports the step at the sign of actual state growth and the first stage of its instrumental-variable version.

Table C.1: The SCE step: robustness

	J (s.e.)	Bootstrap p	N
Panel A. Step J in regression (1)			
Baseline (year effects, separate J_0)	2.815*** (0.506)	0.005	8,028
No year effects	3.201*** (0.523)	0.001	8,028
+ demographics	2.729*** (0.502)	0.003	7,867
+ state fixed effects	2.896*** (0.541)	0.006	8,028
+ state-by-year fixed effects	2.969*** (0.550)	0.010	8,028
+ mood controls (five expectations)	2.478*** (0.512)	0.008	7,594
Excluding symmetric answers	2.626*** (0.498)	0.007	6,326
2014 to 2020 only (dollar answers)	2.603*** (0.776)	0.005	5,496
2022 to 2024 only (direct question)	3.331 (0.636)	0.176	2,532
Excluding round answers	1.835** (0.496)	0.021	4,957
Outcome: national expectation	2.845*** (0.454)	0.003	7,170
Outcome: national, controlling for local	2.499*** (0.461)	0.004	7,170
Threshold estimate (95% interval)	+1.8 [−0.7, +5.0]		8,028
Panel B. Mood placebo: step in other expectations, in standard deviations			
House prices	0.413*** (0.051)	0.001	7,830
Stock prices higher	0.140* (0.052)	0.061	7,818
Gold price	0.001 (0.067)	0.979	7,702
Gas price	−0.011 (0.057)	0.907	7,711
Inflation	0.201* (0.060)	0.069	7,731
Unemployment higher	−0.187** (0.086)	0.014	7,819
Panel C. Actual direction			
Step at the sign of actual state growth	0.824 (1.120)	0.107	8,028
Respondents facing an actual decline			385
First stage: **1**[perceived > 0] on **1**[actual ≥ 0]	0.107 (F = 7.7)		8,028
Two-stage least squares: step at the perceived sign	3.60 (6.29)	0.565	8,028

Notes: Bootstrap p: all 6^{10} Webb sign draws enumerated (fewer with fewer survey years). Demographics: age, income, education, numeracy, tenure. Mood controls: the five other expectations of Panel B. Symmetric answers: today's value midway between the past and expected values. Two-stage least squares: p-value of the reduced form.

D What the National Series Can and Cannot Show

The Michigan step on national growth is identified from the national cycle, 2007–24: one bust and 4 changes of sign. This appendix collects the checks that test it against national conditions; Table D.1 reports every specification.

Macro controls. With unemployment, its change, the mortgage rate, a recession indicator and stock returns, the step is 1.377 (p = 0.007) and the downside slope turns negative. With one bust and correlated cycle variables, these controls cannot fully separate the direction of house prices from other features of 2007–12.

Inflation. With headline CPI inflation as a control, the step falls by about a third and is significant only at the 10 percent level; with the median inflation expectation, it remains significant. With CPI excluding shelter, so that the control does not contain housing costs that follow house prices, the step falls by more than half and is no longer significant. Since excluding shelter makes the step weaker rather than stronger, the drop is not a bad-control effect through rents.

Real expectations. When I subtract each respondent's own one-year inflation expectation from her house-price expectation, and deflate past growth by inflation excluding shelter, the step sits at nominal rather than real zero and is larger. About half of this step, however, comes from the inflation expectation rather than the house-price expectation: inflation expectations fall when house prices turn up. Households that are pessimistic about house prices also expect higher inflation, which points to a general-sentiment component. With CPI excluding shelter as a control, the real-expectations step is no longer significant.

Location. Placebo thresholds give similar steps anywhere from −3 to +1 percent growth (Figure 5), because few months lie near zero.

Whose direction. With one fixed effect per survey month, the regional step is −0.221 ($p = 0.087$), identified from the 18 months in which regions lie on both sides of zero.

The national series therefore shows the pattern but cannot separate it from inflation and sentiment; the SCE, identified within a year, carries the fact.

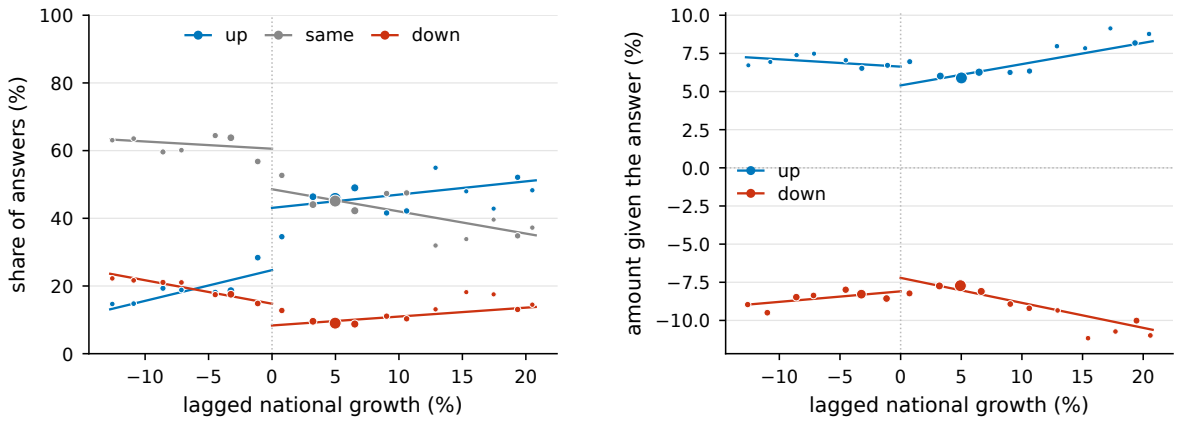
Table D.1: The Michigan step on national growth: controls

	J (s.e.)	Bootstrap p	N
Baseline	1.284*** (0.373)	0.006	89,676
+ demographics	1.209*** (0.354)	0.008	83,896
+ five macro controls	1.377*** (0.376)	0.007	89,676
+ CPI inflation	0.857* (0.430)	0.092	89,676
+ median inflation expectation	1.030** (0.396)	0.016	89,676
+ CPI inflation excluding shelter	0.486 (0.465)	0.377	89,676
Excluding round answers	0.659*** (0.156)	0.001	64,730
Step at real zero	0.563 (0.495)	0.322	89,676
Real outcome (own inflation expectation netted out), step at nominal zero	2.975** (0.743)	0.010	83,874
of which: house-price expectation / inflation expectation	1.355 / −1.620	0.006 / 0.021	83,874
Real outcome + CPI inflation excluding shelter	0.929 (0.685)	0.241	83,874
Regional step / national step, same regression	0.637 / 0.951	0.256 / 0.046	89,676
Regional step with month fixed effects	−0.221* (0.100)	0.087	89,676
Latent step (band model): exact values / round answers as intervals	1.697 / 1.676		91,974
Threshold estimate (95% interval)	+6.9 [−1.5, +7.5]		89,676

Notes: Macro controls: unemployment, its 12-month change, the mortgage rate, a recession indicator, stock returns. Real zero: past growth net of CPI inflation. Real outcome: respondents with an inflation expectation. Latent step: Table E.1; N includes respondents who give a direction without a number.

Shares and amounts. Figure D.1 shows the shares and amounts behind the decomposition of Section 3.7 in Michigan.

Within households. Table D.2 reports the revisions of the reported number among re-interviewed pairs that keep their answer, behind Figure 9. Across all pairs, "same" and "up" answers persist (65.6 and 62.9 percent), while "down" does not (39.2 percent), mostly returning to "same".



(a) Shares of answers **(b)** Amounts given the answer

Figure D.1: Shares and amounts in Michigan, by lagged national growth: shares answering up, same and down, and the average number given by those answering up and down.

Table D.2: Revisions of the reported number among pairs that keep their answer category

	Up / up	Down / down
Pairs	7,846	1,638
Mean first answer	6.60	-9.19
Revision: mean / median	-0.23 / +0.0	+0.50 / +0.0
Identical number (%)	26.1	24.2
Revision on Δs	0.125**	−0.061
	(0.029)	(0.064)
wild cluster bootstrap p	0.036	0.764

Notes: Re-interview pairs, six months apart, answering "up" both times or "down" both times. Revision: second minus first reported percentage change. Δs: change in lagged national growth between the two interviews (pp). OLS with a constant.

E The Band Model

Model. Respondent i in month t holds the latent belief (3), with $\epsilon_{it} \sim N(0,1)$. She reports no change if $|b^*_{it}| < c(s_t)$, a rise if $b^*_{it} > c(s_t)$ and a fall if $b^*_{it} < -c(s_t)$; for given past growth, the band is symmetric around zero, $[-c(s_t), c(s_t)]$, but its width can differ after rises and after declines. Those who report a direction give $y_{it} = b^*_{it} + \eta_{it}$, $\eta_{it} \sim N(0, \tau^2)$. The dispersion and the band take one value after rises and another after declines, $\log \sigma(s) = \log \sigma + \delta_\sigma \mathbf{1}[s < 0]$ and $\log c(s) = \log c + \delta_c \mathbf{1}[s < 0]$; τ is common. In the SCE the latent mean also includes year effects and a separate shift for a perceived change of exactly zero.

Likelihood. Each answer contributes the probability of what the respondent reports. Write μ, σ and c for the latent mean, the dispersion and the band at s_t. An answer of no change contributes the probability that the belief lies inside the band. A rise or a fall reported with a number y contributes the density of y times the probability that the belief lies above, or below, the band given y:

$$\ell_{\text{none}} = \log\left[\Phi\left(\tfrac{c-\mu}{\sigma}\right) - \Phi\left(\tfrac{-c-\mu}{\sigma}\right)\right], \tag{E.1}$$

$$\ell_{\text{rise},y} = \log\left[\tfrac{1}{\omega}\,\phi\left(\tfrac{y-\mu}{\omega}\right)\right] + \log\left[1 - \Phi\left(\tfrac{c-m(y)}{\sqrt{v}}\right)\right], \tag{E.2}$$

$$\ell_{\text{fall},y} = \log\left[\tfrac{1}{\omega}\,\phi\left(\tfrac{y-\mu}{\omega}\right)\right] + \log\Phi\left(\tfrac{-c-m(y)}{\sqrt{v}}\right), \tag{E.3}$$

where $\omega^2 = \sigma^2 + \tau^2$ is the variance of y, and $m(y) = \mu + (\sigma^2/\omega^2)(y - \mu)$ and $v = \sigma^2\tau^2/\omega^2$ are the mean and variance of the belief given y. A rise or a fall reported without a number contributes only the probability that the belief lies above or below the band. The number is not truncated at zero: truncating it would identify only ratios, so dispersion and the band would shrink together.

Identification. The shares of each answer identify the mean and the band relative to dispersion, μ/σ and c/σ. The numbers fix the scale: the location and the total variance $\sigma^2 + \tau^2$. How the numbers behave near the edge of the band separates belief dispersion σ from reporting noise τ.

Estimation. The parameters are estimated by maximum likelihood (BFGS followed by a Nelder–Mead polish), with σ, c and τ in logs; ten random starts reach the same maximum. Standard errors are clustered by year, and intervals for the latent step and the bands come from a year bootstrap with 999 draws. The table that follows reports the estimates for the SCE, for renters and owners separately, and for Michigan. The belief rule (11) of the life-cycle model uses the SCE estimates: the mean latent belief just below and just above a perceived change of zero is 0.386 and 3.306 percent.

Table E.1: The band model: SCE (perceived local change) and Michigan (national and regional growth)

	SCE			Michigan	
	All	Renters	Owners	National	Regional
Latent step J	2.920***	4.746***	2.133***	1.697**	2.324***
	[1.76, 3.84]	[2.40, 7.83]	[0.97, 2.97]	[0.43, 2.68]	[1.30, 3.45]
Shift at a perceived zero, J_0	3.013***	4.919***	2.030***	—	—
	[1.61, 4.20]	[2.14, 8.18]	[0.92, 2.92]		
Slope below zero, β^-	−0.100	−0.218	−0.017	0.158***	0.002
	[-0.35, 0.13]	[-0.61, 0.15]	[-0.20, 0.18]	[0.08, 0.52]	[-0.08, 0.12]
Slope above zero, β^+	0.181***	0.201***	0.167***	0.031	0.065
	[0.14, 0.24]	[0.14, 0.28]	[0.11, 0.24]	[-0.05, 0.14]	[-0.05, 0.16]
Band after rises, c^+	1.24	1.60	1.15	3.39	3.41
	[1.17, 1.31]	[1.50, 1.73]	[1.08, 1.24]	[3.16, 3.64]	[3.19, 3.67]
Band after declines, c^-	1.36	1.76	1.27	4.68	4.70
	[1.18, 1.50]	[1.37, 2.30]	[1.09, 1.41]	[4.42, 4.86]	[4.42, 4.87]
Dispersion after rises, σ^+	7.58	9.67	6.53	5.60	5.67
	[6.70, 8.49]	[8.59, 10.70]	[5.74, 7.38]	[4.97, 6.33]	[5.04, 6.37]
Dispersion after declines, σ^-	9.68	11.73	8.72	6.08	6.02
	[8.89, 10.65]	[10.15, 13.38]	[7.82, 9.55]	[5.61, 6.57]	[5.61, 6.53]
Reporting noise, τ	0.76	1.13	0.68	3.72	3.70
	[0.69, 0.83]	[0.97, 1.39]	[0.61, 0.74]	[3.30, 4.10]	[3.29, 4.08]
$p(c^- = c^+)$	0.142	0.460	0.178	0.001	0.001
Mean latent belief just below zero	0.386	−0.533	0.931	—	—
Mean latent belief just above zero	3.306	4.213	3.064	—	—
Renters minus owners, latent step	2.613 [0.40, 6.02], $p = 0.018$				
N	8,028	2,059	5,903	91,974	91,974

Notes: 95% intervals and stars from a bootstrap that resamples survey years (999 draws). Michigan: months of the main sample; N exceeds that of Table 3 because the 2,298 respondents who give a direction without a number enter through their category. Renters minus owners: the same difference in answers has a wild cluster bootstrap p-value of 0.120.

F The Two-Period Model

The consumption responses and the gain from buying now are derived in Section 4.1; this appendix derives the welfare cost of a belief error. Under the true g,

$$
\begin{aligned}
V(a) &= \ln(y_0 - a) + \beta \ln\left(y_1 + Ra + (1+g)p_0 H\right), \\
V'(a) &= -\frac{1}{c_0} + \frac{\beta R}{c_1} = 0 \quad \text{at } a^*, \\
V''(a^*) &= -\frac{1}{c_0^2} - \frac{\beta R^2}{c_1^2} = -\frac{1+\beta}{\beta c_0^2} \quad \text{using } c_1 = \beta R c_0.
\end{aligned}
$$

Acting on $\hat{g}$ changes saving by $\Delta a \equiv \hat{a} - a^* = (g - \hat{g})\, p_0 H/[(1+\beta)R]$. A second-order expansion around a^* gives the loss

$$
L \equiv V(a^*) - V(\hat{a}) \approx \tfrac{1}{2}\,|V''(a^*)|\,(\Delta a)^2 = \frac{1}{2} \cdot \frac{1+\beta}{\beta c_0^2} \cdot \frac{(g-\hat{g})^2\, p_0^2 H^2}{(1+\beta)^2 R^2} = \frac{h^2(\hat{g}-g)^2}{2\beta(1+\beta)}, \qquad h \equiv \frac{p_0 H}{R c_0}.
$$

Cutting consumption by a fraction λ in both periods costs $-(1+\beta)\ln(1-\lambda) \approx (1+\beta)\,\lambda$, so $\lambda \approx L/(1+\beta)$, which is (5).

G The Life-Cycle Model

Solution. The model is solved by the discrete-continuous endogenous grid method with extreme-value taste shocks on the tenure choice (Iskhakov et al., 2017); it agrees with value function iteration on the same problem, and median Euler-equation errors are of the order of 10^{-5}. Price risk is integrated with three Gauss–Hermite nodes.

Simulation. The population alive in 1998 comes from the steady state of the calibrated economy; a new cohort enters at 25 each year from 1999, reusing the age-25 wealth and income draws of the steady state, with its own later income and taste shocks. Both households receive the same draws. Prices follow actual Case-Shiller December-to-December growth through 2025.

Robustness. Table G.1 describes the marginal renters (Panel A), reports every version of the comparative statics (Panel B) and reports the welfare losses (Panel C). Two rows of Panel B change the environment rather than a parameter. Giving renters and owners their own belief rules, from the SCE band model by tenure, the wave of purchases still comes in 2012. Replacing the fixed price-to-rent ratio by rent following the CPI rent index, the timing at the turn is unchanged, but step households, who expect prices to grow no faster than rents, own too little during the boom.

Table G.1: The life-cycle model: marginal renters, comparative statics and welfare

Panel A. Renters in 2011 by what they do under each belief

	Mistimers	Late buyers	Non-buyers
Share of renters (%)	17.9	28.7	37.0
Age	47.0	56.5	42.3
Income relative to the mean at that age	0.84	0.81	0.51
Liquid wealth over the down payment, median	1.58	2.76	0.12
Gain from buying in 2011 under own belief (median, % of income)	−9.7	−40.7	−125.7
Gain from buying in 2011 under the true path (median, % of income)	19.2	20.9	−57.1
Spending response per point of ten-year expected growth (%)	−9.80	−6.10	−1.44

Panel B. Comparative statics

	Step household buying in 2012 (%)	Rise from 2010 (pp)	Full information buying in 2011 (%)
Baseline	24.9	13.7	78.9
Loan-to-value limit 0.80	19.2	9.8	70.1
Loan-to-value limit 0.95	24.9	14.6	83.3
Transaction cost 2.5%	34.0	19.2	83.7
Transaction cost 10%	14.6	6.6	68.4
Step at the low end of its interval ($J = 1.76$)	15.3	3.6	78.8
Step at the high end of its interval ($J = 3.84$)	33.9	22.5	78.9
Taste-shock scale 0	25.6	14.2	79.9
Taste-shock scale 0.02	25.3	14.7	79.8
Taste-shock scale 0.08	26.7	12.4	74.6
Renter-specific belief rules	31.6	25.2	78.9
Rent from the CPI rent index	23.0	12.0	77.9

Panel C. Welfare loss of the step household (% of lifetime consumption)

	Mean	Median
All households	5.31	2.58
Without the bequest term	4.95	2.84
Against full information with the step household's price risk	5.22	2.46
Without the ownership premium	4.79	2.51
Renter-specific belief rules	5.15	2.66
Rent from the CPI rent index	3.03	1.27
Share of the total loss from households whose tenure path differs (%)	96.5	

Notes: Mistimers buy in 2012 under the step belief and in 2011 under full information; late buyers buy in 2011 under full information but in neither year under the step belief; non-buyers buy in neither year under either belief. Panel B: purchases peak in 2012 for the step household and in 2011 under full information in every row.